

Chapter 4

Brushless Motor Fundamentals

The design of brushless permanent magnet motors is not a simple task. On a more general level, motor design requires knowledge of magnetism, mechanics, thermodynamics, electronics, acoustics, and material science. On a more specific level, it requires knowledge of performance requirements and constraints imposed by the intended motor application. Given this body of knowledge, motor design involves finding an optimal solution for the least cost. This text focuses on the magnetic aspects of motor design. Other general aspects listed above are considered in the design process, but detailed design information about other areas is not provided.

4.1 Assumptions

Besides the performance requirements discussed above, other initial assumptions are necessary to more clearly define and focus the initial design of brushless permanent magnet motors. Some of these assumptions add restrictions and others identify conventional design techniques.

Rotational Motion

It is assumed that rotary motion is desired. While the design techniques developed here are easily applied to motors having linear motion, initial work will focus on rotary motion where the rotor is inside a stator.

Surface-Mounted Magnets

Most brushless permanent magnet motors have magnets mounted on the rotor surface facing an air gap. For this reason, initial work will focus on this topology. In some motors the permanent magnets are buried within steel structures. Interior permanent magnet topologies generally find application for three reasons. First, by burying magnets, it is possible to employ flux concentration. Second, enclosing magnets in steel can make the rotor structurally stronger and therefore allow operation at higher speeds. Finally, by burying magnets it is possible to drive a motor over a wider speed range through the use of field weakening control.

4.2 Fundamental Concepts

In brushless motor design, mutual torque and back EMF are the two fundamental parameters to be determined. These two parameters are intimately linked through (3.43), so knowledge of one provides information about the other. While the BLi and BLv laws can be used to determine the torque and back EMF respectively, it is more convenient to compute flux linkage, and then employ Faraday's law to obtain the back EMF. Then, (3.43) can be used to determine torque.

Magnetic Circuit Model

Consider the motor cross section shown in Fig. 4-1. Here the rotor contains $N_m=4$ magnet poles facing the air gap. As a result there is a factor of two difference between electrical and mechanical measures, $\theta_e=(N_m/2)\theta_m$. For simplicity the stator is shown without slots or windings. The magnet flux leaving North poles at the air gap crosses over to the stator and splits into two equal sections, each traveling in the opposite direction and crossing the air gap toward South poles at the air gap. For one half of a North and South pole facing the air gap, this flux flow is illustrated by the flux path on the right side of the figure. Flux flowing between each of the other adjacent half pole pairs follows accordingly.

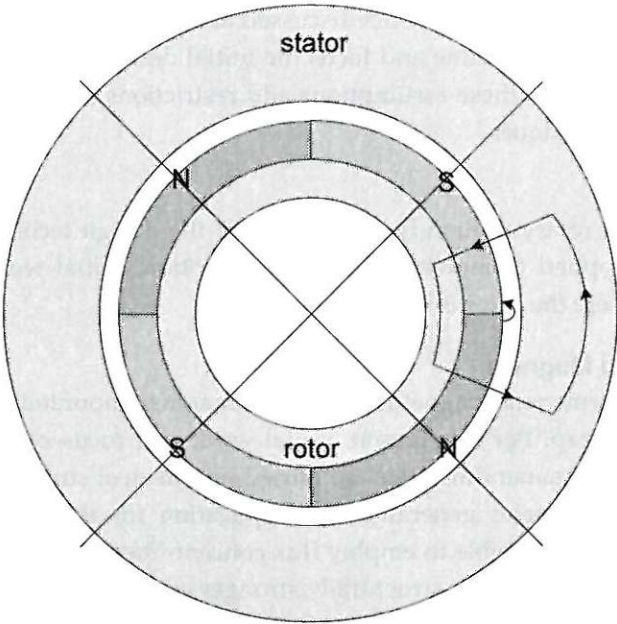


Figure 4-1. Fundamental motor structure and associated flux paths.

In addition to the primary flux path shown, some magnet flux jumps from one magnet to the next in the air gap without passing into the stator, as illustrated by the path in the air gap on the right of Fig. 4-1. The flux that follows this path is often called *magnet leakage flux*.

Because the flux paths shown in Fig. 4-1 repeat for every adjacent half pole pair, it is only necessary to model one such pair as shown in Fig. 4-2. In this figure, the rotor and stator steel areas are modeled simply as reluctances R_r and R_s respectively. The two half magnets are modeled as a flux source ϕ_r and associated magnet reluctance R_m , with the direction of the flux source dictating the magnet polarity. Primary flux flow from the magnets across the air gap into the stator flows through the air gap reluctances denoted R_g . Leakage flux from one magnet to the next flows through the leakage reluctance R_l . The three circuit fluxes are the magnet flux ϕ , the air gap flux ϕ_g , and the leakage flux ϕ_l .

Before determining a back EMF, the magnetic circuit must be solved to determine the air gap flux density B_g . Rather than solving the magnetic circuit as shown in Fig. 4-2, it is convenient to simplify the circuit as shown in Fig. 4-3. Since the right magnet and the rotor reluctance are in series, they are swapped in Fig. 4-3a. This places the two half magnets next to each other and places the rotor reluctance next to the other reluctances. At this point it is difficult to determine an analytical description of the leakage reluctance. However, the percentage of flux that travels the primary flux path

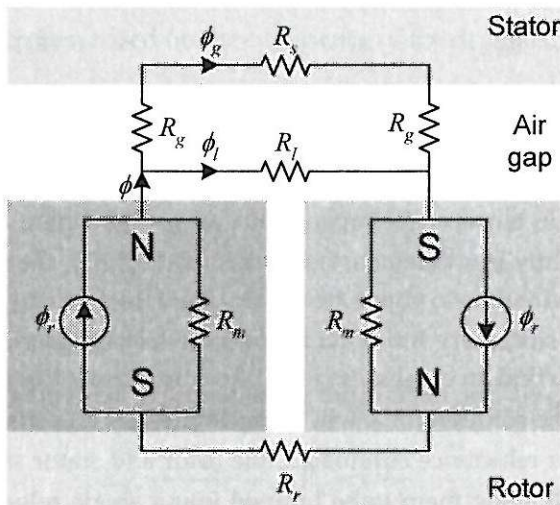


Figure 4-2. A magnetic circuit model for the structure shown in Fig. 4-1.

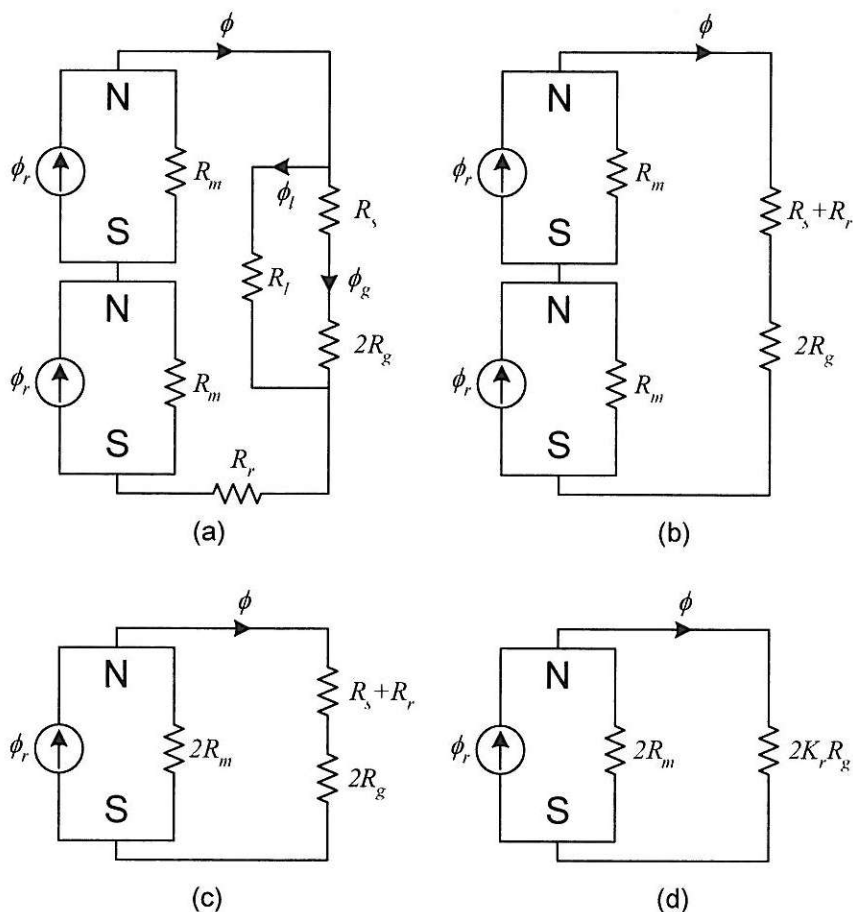


Figure 4-3. Simplifications of the magnetic circuit in Fig. 4-2.

across the air gap relative to the magnet flux can be estimated. That is, the air gap flux can be written in terms of the magnet flux as $\phi_g = K_l \phi$, where K_l is a *leakage factor* that is typically slightly less than one. Using this relationship, the next step in simplifying the magnetic circuit is to eliminate the leakage reluctance R_l as shown in Fig. 4-3b. This is possible since very little flux follows the leakage path, and it is desirable since it is difficult to find an expression for R_l . To compensate for the flux that follows this leakage path, the solution for ϕ will be multiplied by an estimate of K_l to obtain ϕ_g . With the leakage reluctance eliminated, the rotor and stator steel reluctances are in series, thereby allowing them to be lumped into a single reluctance as shown in Fig. 4-3b as well.

The two magnet halves in series in Fig. 4-3b can be simplified as shown in Fig. 4-3c. From an electrical circuit point of view, the simplified magnet shown in Fig. 4-3c is found by determining the Norton equivalent circuit of the two series magnet halves. That is, the simplified flux source is ϕ_r since that is the flux that would flow if a "short" were placed across the series magnets, and $2R_m$ is the equivalent reluctance seen looking into the circuit formed by the two series magnets. From a magnetic material point of view, the two half magnets in series is equivalent to a single block of permanent magnet material having twice the length. Therefore, ϕ_r remains unchanged but R_m doubles since reluctance is directly proportional to material length.

The steel reluctance $R_r + R_s$ in Fig. 4-3c is nonlinear because of the saturation characteristic of ferromagnetic materials. Therefore, this reluctance must be eliminated in some way to find an analytic solution. As long as the permeability of the steel is high relative to air, the steel reluctance will be small relative to the air gap reluctance R_g as demonstrated in the example in Chapter 2. When this is true, the steel reluctance can be thought of as a perturbation of the air gap reluctance. That is, the steel reluctance can be eliminated by introducing a *reluctance factor* K_r as shown in Fig. 4-3d. Here K_r is a constant slightly greater than one that increases the air gap reluctance slightly to accommodate or compensate for the missing steel reluctance.

It is important to note that in practice one seldom tries to determine analytical expressions for the leakage factor K_l and reluctance factor K_r . It is simply too difficult to determine accurate values given the simple modeling being performed here. Their values are usually chosen based on the experience of the designer.

Magnetic Circuit Solution

Given the magnetic circuit in Fig. 4-3d, the magnet flux ϕ can be expressed using flux division as (*i.e.*, as in current division between resistors in an electrical circuit)

$$\phi = \frac{2R_m}{2R_m + 2K_r R_g} \phi_r = \frac{1}{1 + K_r \frac{R_g}{R_m}} \phi_r \quad (4.1)$$

Based on $\phi_g = K_l \phi$, and general expressions for the magnet and air gap reluctances, *i.e.*,

$$R_m = \frac{l_m}{\mu_R \mu_0 A_m}, \quad R_g = \frac{g}{\mu_0 A_g} \quad (4.2)$$

the air gap flux can be written as

$$\phi_g = K_I \phi = \frac{K_I}{1 + K_r \frac{\mu_R g A_m}{l_m A_g}} \phi_r \quad (4.3)$$

where l_m and A_m are the magnet length and cross-sectional area respectively, and g and A_g are the air gap length and cross-sectional area respectively. Substituting the flux concentration factor $C_\phi = A_m/A_g$ from (2.30), the flux density relationships $B_g = \phi_g/A_g$ and $B_r = \phi_r/A_m$, and the permeance coefficient from (2.33) as $P_c = l_m/(gC_\phi)$ into (4.3) gives an air gap flux density of

$$B_g = \frac{K_I C_\phi}{1 + K_r \frac{\mu_R}{P_c}} B_r \quad (4.4)$$

This equation describes the air gap flux density crossing the air gap. For the motor being considered here with surface magnets, the leakage factor is typically in the range $0.9 \leq K_I < 1.0$, the reluctance factor is in the range $1.0 < K_r \leq 1.2$, and the flux concentration factor is ideally 1.0. If one considers these values to be fixed and the remanence B_r to be fixed by the magnet choice, the permeance coefficient P_c determines the amplitude of the air gap flux density. As the permeance coefficient increases, the air gap flux density approaches a maximum that is slightly less than the remanence. Without flux concentration, it is not possible to achieve an air gap flux density B_g greater than B_r . Moreover, the relationship between permeance coefficient and air gap flux density is nonlinear. The air gap flux density approaches the remanence asymptotically. Doubling P_c does not double B_g . However, doubling P_c means doubling the magnet length, which doubles its volume and associated cost. For typical parameter values, Fig. 4-4 demonstrates the relationship between permeance coefficient and the ratio B_g/B_r , where the vertical lines mark the typical four to six permeance coefficient range used in many motor designs.

The flux density in (4.4) defines an approximation to the air gap flux density over the surface of the magnet pole. That is, (4.4) gives the amplitude of the air gap flux density $|B_g|$ as shown in Fig. 4-5. Over North poles, (4.4) gives the positive amplitude, and over South poles, (4.4) gives the negative amplitude. While this approximation is far from exact, the derivation of (4.4) provides valuable insight into motor operation, and (4.4) itself illustrates fundamental principles that exist even when more accurate modeling is performed.

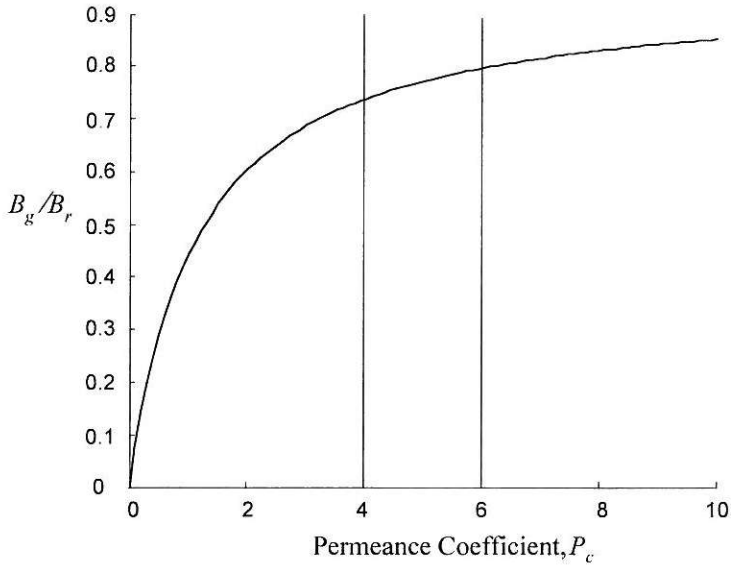


Figure 4-4. Relationship between normalized air gap flux density and permeance coefficient.

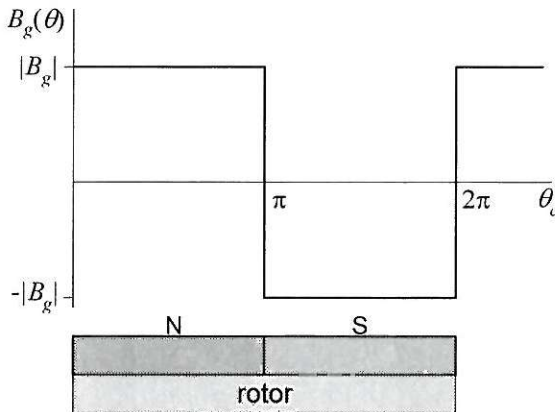


Figure 4-5. Ideal air gap flux density distribution.

For convenience, the horizontal axis in Fig. 4-5 is described in terms of electrical measure, which is periodic with respect to one pair of poles as shown in the figure. For the prototype motor being considered here, there are two electrical periods around the circumference of the rotor.

Flux Linkage

Given the solution of the magnetic circuit, consider the addition of two slots containing a winding composed of N turns of wire as shown in Fig. 4-6. The winding forms a coil that comes out of the slot at the top of the figure and goes into the slot on the right. The *coil pitch* or *coil throw* is 180°E or $\theta_p = 2\pi/N_m \text{ radM}$, which is said to be a *full pitch* winding. As the rotor turns, the air gap flux links this coil. For the rotor position shown in Fig. 4-6, the flux flows toward the South pole of the rotor magnet across the air gap from the coil. This flux flow direction is the opposite of that produced by current flowing in the coil, so the flux linkage is negative. If ϕ_g is the air gap flux as given in (4.3), the flux linkage at this position, designated $\theta_e = 0$, is $\lambda = -N\phi_g$.

If the rotor turns 90°E as shown in Fig. 4-7, the coil is centered over one half of a South pole and one half of a North pole. Over the South pole the flux flows toward the rotor, while over the North pole the flux flows away from the rotor. The net flux linked by the coil is the sum of these two components, which is zero.

Rotating the rotor another 90°E to a position $\theta_e = 180^\circ\text{E}$, as shown in Fig. 4-8, the coil is now centered over a North pole. The flux linked at this position is equal in magnitude to the flux linked at the $\theta_e = 0$ position shown in Fig. 4-6, but the direction is opposite. Therefore, the flux linkage is positive.

At intermediate points between 0°E and 180°E the flux linkage varies linearly between the minimum at 0°E and the maximum at 180°E . Likewise, as the rotor

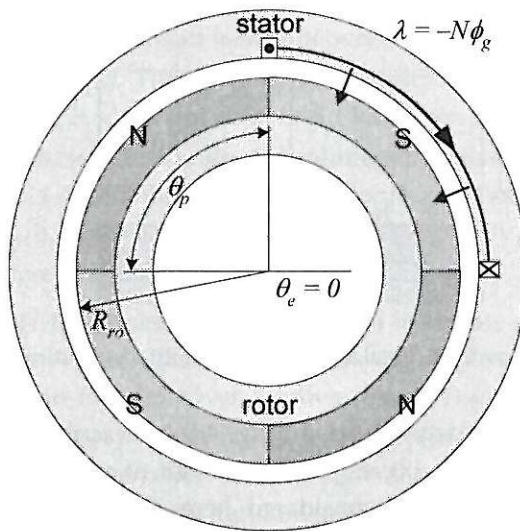


Figure 4-6. Motor having one full-pitch coil.

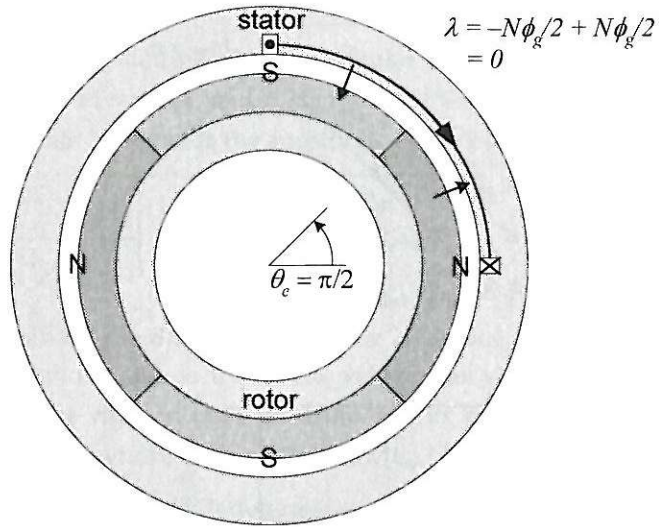


Figure 4-7. Motor with rotor at 90°E.

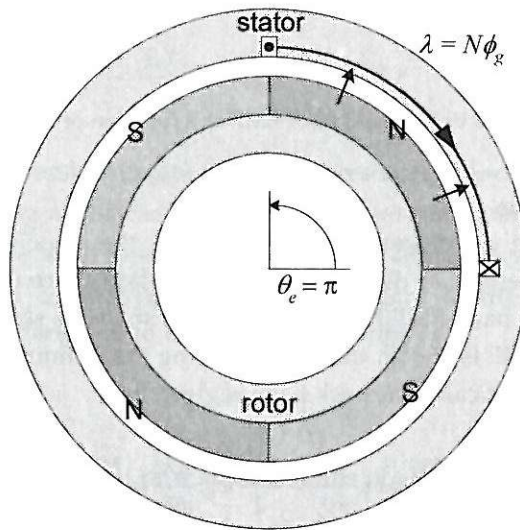


Figure 4-8. Motor with rotor at 180°E.

rotates past 180°E toward 360°E, the flux linkage falls linearly from the maximum at 180°E back toward another minimum at 360°E. Further rotor rotation creates a periodic flux linkage waveform as South and North poles alternately link the coil as shown in Fig. 4-9a.

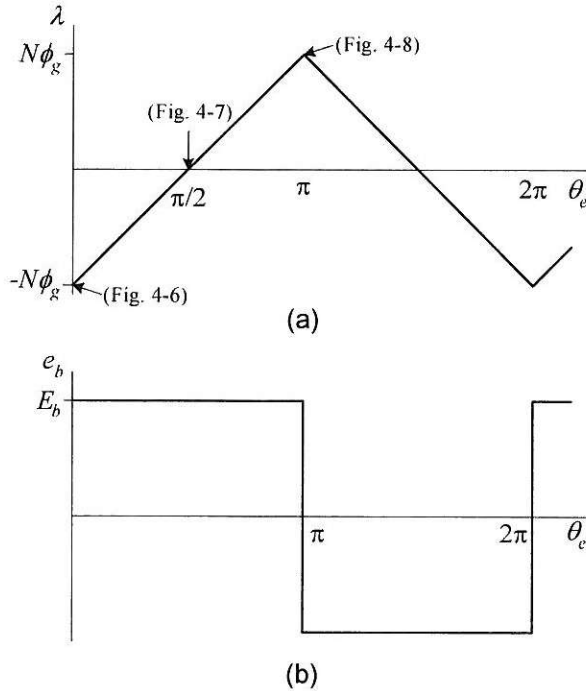


Figure 4-9. Flux linkage and back EMF as a function of rotor position.

Back EMF and Torque

From the flux linkage waveform shown in Fig. 4-9a, the associated back EMF is the derivative of the waveform as dictated by Faraday's law. Since the flux linkage is triangular in shape, the back EMF has a square wave shape as shown in Fig. 4-9b. The period of the back EMF is 360° , thereby justifying the definition of electrical measure in Chapter 1. Analytically, the back EMF is given by

$$e_b = \frac{d\lambda}{dt} = \frac{d\theta_e}{dt} \frac{d\lambda}{d\theta_e} = \omega_e \frac{d\lambda}{d\theta_e} = \frac{N_m}{2} \omega_m \frac{d\lambda}{d\theta_e} = \frac{N_m}{2} \omega_m \frac{2N\phi_g}{\pi} \quad (4.5)$$

where ω_m is the rotor speed in rad/s. To simplify this further, the air gap flux can be written as

$$\phi_g = B_g A_g = B_g R_{ro} \theta_p L_{st} = \frac{2\pi}{N_m} B_g L_{st} R_{ro} \quad (4.6)$$

where B_g is the air gap flux density (4.4) as modified by Carter's coefficient (2.13) or (2.14) to take into account the slotting, θ_p is the angular pole pitch in radM, R_{ro} is the air gap radius at the magnet surface, and L_{st} is the axial length of the motor. Substituting this relationship into (4.5) yields the amplitude of the back EMF E_b

$$|e_b| = E_b = \frac{N_m}{2} \omega_m \frac{2N}{\pi} \left(\frac{2\pi}{N_m} B_g L_{st} R_{ro} \right) = 2NB_g L_{st} R_{ro} \omega_m = K_e \omega_m \quad (4.7)$$

This expression agrees with the BLv law. The factor $2N$ is due to the two slots each having N conductors, and $R_{ro}\omega_m$ is the linear velocity at which the flux linkage changes. As shown on the right in (4.7), all terms in this expression except for ω_m form a *back EMF constant* K_e , whose units are V/(radM/s).

Application of (3.43) to (4.5) and (4.7) determines the torque produced by a current i flowing in the coil. Because (3.43) is a simple algebraic relationship, for constant current the torque versus position shape is the same as that for the back EMF versus position as shown in Fig. 4-9b. The amplitude of the torque is given by

$$|T| = \frac{E_b i}{\omega_m} = 2NB_g L_{st} R_{ro} i = K_t i \quad (4.8)$$

All terms in this expression except for R_{ro} represent the force experienced by the rotor. This force acting at the radius R_{ro} gives the torque according to (1.1). From a different point of view, all terms in (4.8) except for i , form a *torque constant* K_t , whose units are N·m/A. By comparing this torque constant to the back EMF constant described earlier, they are seen to be the same quantity, *i.e.*,

$$K_e = K_t = 2NB_g L_{st} R_{ro} \quad (4.9)$$

and they are numerically equal provided the units of all terms are interpreted consistently, as they are in SI units.

The flux, flux linkage, back EMF, and torque described in this section represent an ideal situation. In reality, the air gap flux density does not have a square wave shape as shown in Fig. 4-5. As a result, the flux linkage does not exhibit the ideal triangular shape shown in Fig. 4-9a, and so the back EMF is not a square wave as shown in Fig. 4-9b. Much more rigor is required to accurately determine these waveforms. At the same time, the preceding analysis provides significant insight into motor operation.

Two more common and more realistic sets of waveforms are shown in Fig. 4-10. Because of flux leaking from magnet to magnet, motors having full pitch windings like those considered in this chapter typically exhibit a more trapezoidal back EMF waveform as shown by the lighter curves in the figure. Other motors are designed so that the flux linkage, back EMF, and torque are sinusoidal, which are shown by the darker curves in the figure. These motors typically do not have full pitch windings.

Multiple Coils

The motor shown in Figs. 4-6 through 4-8 is not very efficient because room exists for more coils. The flux from three magnet poles goes unused at all times. To increase the motor performance, three more full pitch coils can be added as shown in Fig. 4-11. In

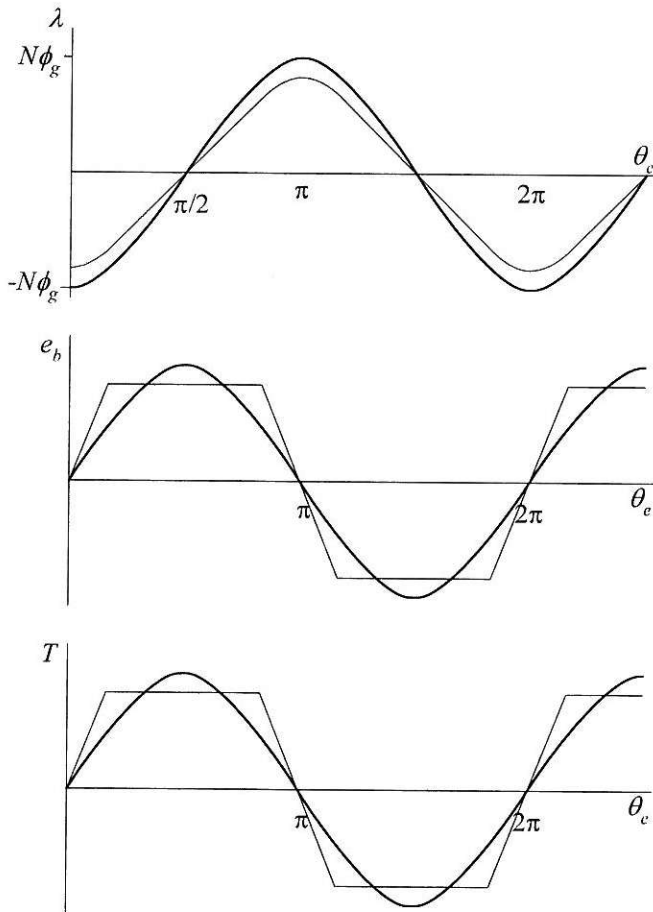


Figure 4-10. More typical motor waveforms.

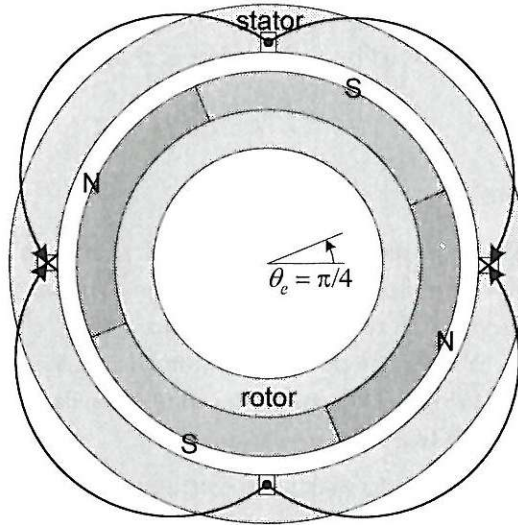


Figure 4-11. A motor containing four full-pitch coils.

this figure, two more slots are added, making room for the new three coils. Now each slot contains two coil sides rather than one. The first coil remains unchanged. Starting with the first coil and moving counterclockwise around the stator, each successive coil is wound in the opposite direction as the last. In this way, the flux linked to each coil is the same as the preceding coil because the North and South magnet poles alternate as well.

At this point there is flexibility about how the individual coils are connected. With four coils, several possibilities exist. In all cases, the collection of connected coils is called a *phase winding* or simply a *phase*.

In most cases, all coils in a phase are connected in series. That is, the end of one coil is connected to the start of the next coil. When this is done the back EMFs of each coil add together to become the net back EMF for the entire winding. Since the back EMF for each coil has an identical shape, the back EMF amplitude (4.7) simply changes to

$$E_b = 2N_m NB_g L_{st} R_{ro} \omega_m \quad (4.10)$$

where $N_m=4$ in this example. Similarly, the amplitude of the torque scales by N_m ,

$$|T| = 2N_m NB_g L_{st} R_{ro} i \quad (4.11)$$

This expression confirms that $T=kD^2L$ as given by (1.8). One diameter D appears as the radius R_{ro} ; the other is implied by the number of magnet poles N_m , and L appears directly as L_{st} .

4.3 Multiple Phases

The motor considered in the preceding section and shown in Fig. 4-11 is a single phase motor. This motor type does not appear in many applications because it is not possible to produce torque at all rotor positions. Once every 180°E the back EMF and torque cross through zero. At these points, the motor cannot produce torque. Moreover, if the motor comes to rest at these points, the motor cannot be started without physically rotating the shaft to a nonzero torque point.

To eliminate this problem and to make it easier to produce constant torque, brushless permanent magnet motors contain more than one phase winding, with the individual phase windings oriented so that their zero crossing points in the back EMF and torque are uniformly distributed over an electrical period. Most brushless permanent magnet motors have three phases, and some have either one or two phases. Because the number of power electronics devices needed to drive a motor increases with the number of phases, it is very uncommon to see a motor having more than three phases. Only at very high power levels where multiple banks of power electronic devices are required does it make sense to use more than three phases.

The motor shown in Fig. 4-11 can accommodate two additional phase windings, each composed of four coils in the same way that the first phase winding has four coils. The resulting three phase motor is shown in Fig. 4-12, where A, B, and C designate the phases. To avoid making the figure visually confusing, Fig. 4-12 shows only one coil for each phase. The rest are identified by the phase labels near each slot.

The phase A winding shown in Fig. 4-12 matches the phase winding described earlier in Fig. 4-11. Since there are three phases, the zero crossings of the back EMF and torque are separated by $(360^\circ\text{E})/3$, which is 120°E or 60°M . Therefore the slots for phase B are rotated 60°M from the corresponding slots of phase A, and the slots of phase C are rotated 60°M from the corresponding slots of phase B. This arrangement places slots around the circumference of the stator every 30°M as shown in the figure. The resulting phase waveforms are shown in Fig. 4-13. Since the rotor is unchanged, the flux linkage, back EMF, and torque of phase B will have the same shape as corresponding phase A waveforms, but will be delayed by the 120°E offset due to the 120°E offset in slot placement. Likewise phase C waveforms look like phase A waveforms displaced by 240°E .

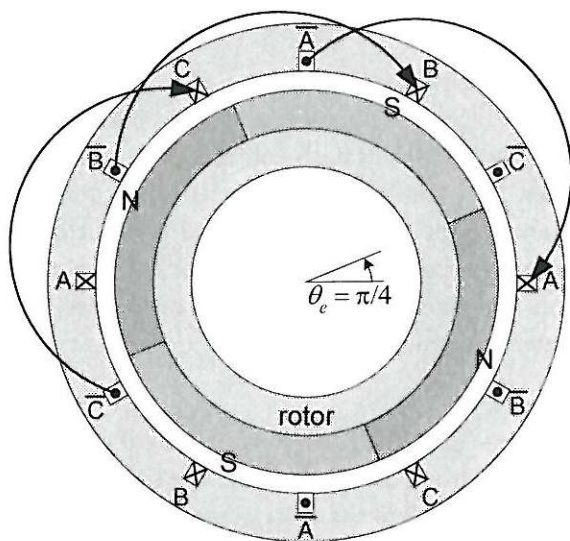


Figure 4-12. A three phase motor.

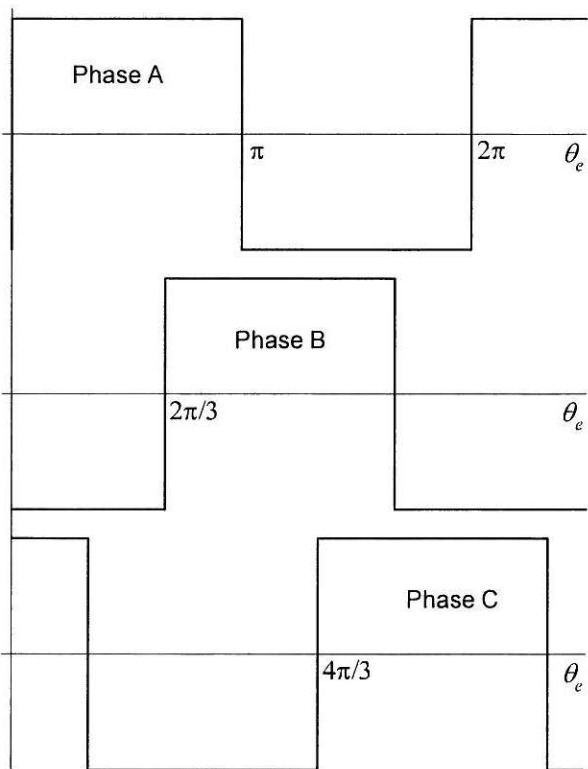


Figure 4-13. Back EMF and torque waveforms for a three phase motor.

4.4 Design Variations

The motor considered in the preceding section had full pitch coils as well as magnet poles that spanned 180°E . In addition, all four coils in each phase were aligned with each other. That is, the back EMFs of each coil of a given phase were in phase alignment with one another. As a result, when the coils were connected in series, the net back EMF had the same shape as that for a single coil but had an amplitude four times larger. While these properties make the motor relatively easy to analyze, relatively few real motors are constructed with these properties. In this section, variations in these parameters are studied.

Fractional Pitch Coils

When the coil pitch differs from 180°E , the winding is called a *fractional pitch winding*. To see how a fractional pitch winding influences the back EMF and torque, consider the motor shown in Fig. 4-14. Here the rotor remains unchanged, but the coil spans an angle of $\theta_c=120^\circ\text{E}$, called the angular coil pitch, where the *coil pitch factor* is defined as $\alpha_c=\theta_c/\theta_p$. At the initial position $\theta_e=0$ shown in the figure, the flux linkage is smaller in amplitude than the full pitch case by the coil pitch factor.

When the rotor rotates to $\theta_e=60^\circ\text{E}$ or $\pi/3\text{radE}$ as shown in Fig. 4-15, the flux linkage increases from the minimum to zero since an equal amount of flux travels both ways through the coil. When the rotor reaches $\theta_e=120^\circ\text{E}$ or $2\pi/3\text{radE}$ as shown in Fig. 4-16,

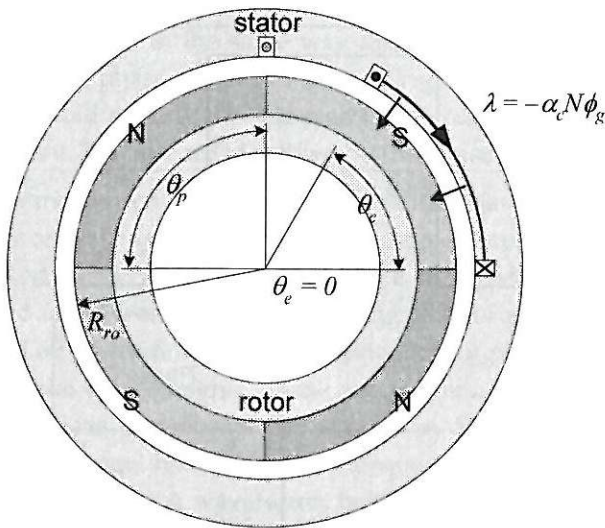


Figure 4-14. Motor having one fractional pitch coil.

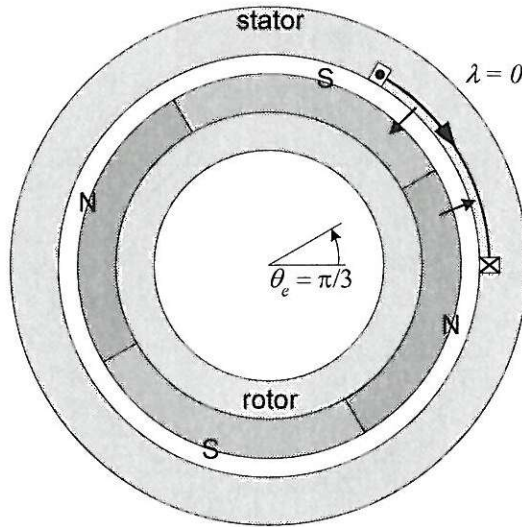


Figure 4-15. Motor with rotor at 60°E.

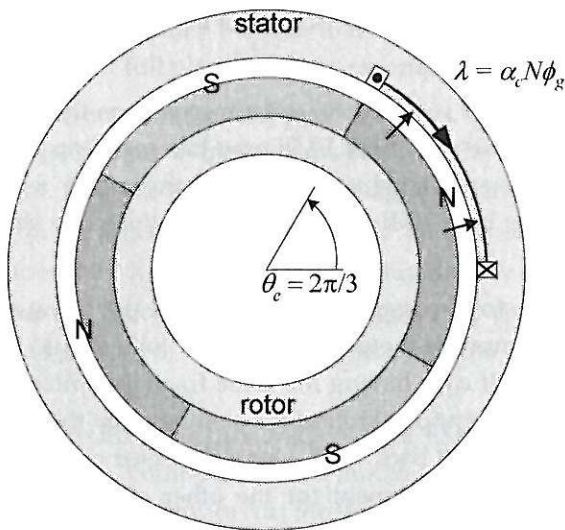


Figure 4-16. Motor with rotor at 120°E.

the flux linkage reaches a maximum since the coil faces only the North magnet pole. In addition, the flux linkage remains at the maximum until $\theta_e = 180^\circ\text{E}$, whereupon it starts decreasing through zero to the minimum again. Continuing to rotate the rotor allows the flux linkage waveform to become known. Figure 4-17 shows the flux link-

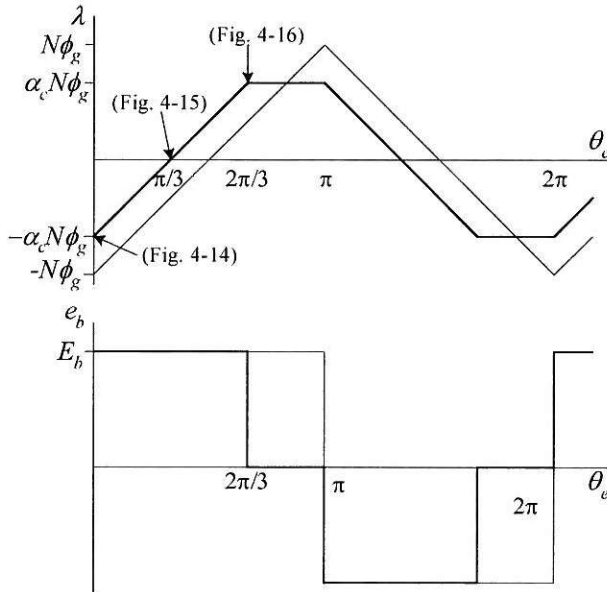


Figure 4-17. Flux linkage and back EMF for the fractional pitch coil case.

age and resulting back EMF for this case superimposed on the associated waveforms for the ideal full pitch case. The back EMF now has segments over which it is zero. These zero segments are associated with position ranges over which the flux linkage does not vary. Since the back EMF and torque have the same shape, the torque has zero segments as well.

As before, three of the four magnets in this motor are not being used for torque production. Therefore, to improve motor performance, coils should be added to interact with the other magnets. If coils having the same fractional pitch are used, they must be placed in appropriate locations around the stator so that the back EMF of each coil adds constructively to the net back EMF. Using the rotor pole transitions as a guide, Fig. 4-18 shows a possible placement for the other coils. One side of each coil is aligned with a magnet pole transition, and every other coil is wound in the opposite direction. If all coils are connected in series to form a phase winding, the back EMF will have the same shape as that shown in Fig. 4-17, but will have an amplitude N_m times larger.

Fractional Pitch Magnets

Magnets seldom span a full pole pitch of 180° because flux at the transitions between North and South poles does not contribute to torque, but rather travels

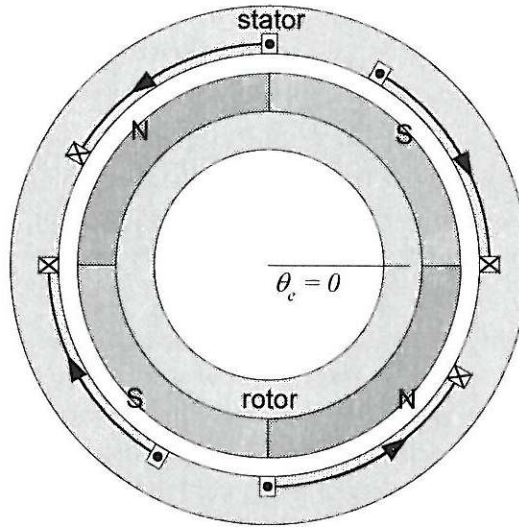


Figure 4-18. A motor containing four fractional pitch coils in one phase.

directly from one pole to the other without linking to coils in stator slots. As such, magnet material is wasted if full pitch magnets are employed.

To understand the impact of fractional pitch magnets, consider the full pitch coil motor shown in Fig. 4-19. Here, there are gaps between the magnet poles containing nonmagnetic material. Therefore, the angular magnet pitch θ_m is smaller than the angular pole pitch θ_p . As a result, the magnet and air gap cross-sectional areas are equal to $R_{ro}\theta_m L_{st}$ rather than $R_{ro}\theta_p L_{st}$. The air gap flux density expression in (4.4) still applies, but describes the flux density in the air gap over the magnet surface only. Under the conditions assumed here, no flux crosses the air gap over the gaps between the magnets.

The procedure for computing the flux linkage, back EMF and torque is unchanged. For any given rotor position, compute the flux linked to the stator coil. By identifying key positions where flux transitions occur, the flux linkage can be plotted. Differentiating the flux linkage gives the back EMF, which has the same shape as the torque produced by the coil under constant current conditions. Performing these steps for the motor shown in Fig. 4-19 leads to the waveforms shown in Fig. 4-20, where the waveforms for this case are superimposed on the associated waveforms for the ideal full pitch magnet and coil case.

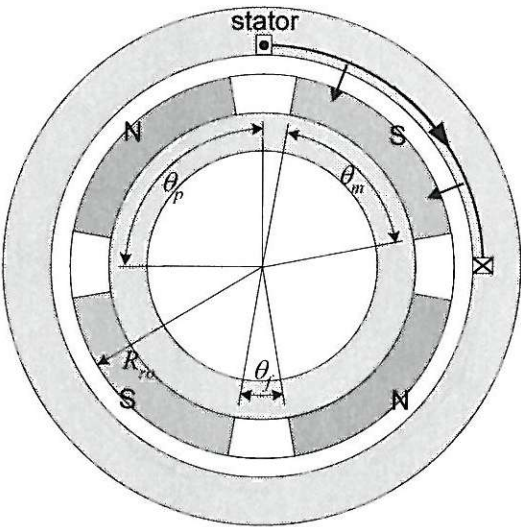


Figure 4-19. A motor having fractional pitch magnets.

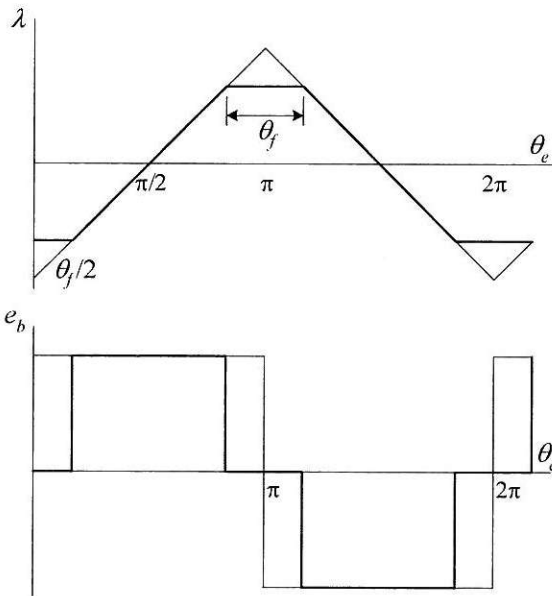


Figure 4-20. Flux linkage and back EMF for the fractional pitch magnet case.

Fractional Slot Motor

The motor shown in Fig. 4-12 has $N_s=12$ slots, $N_m=4$ magnet poles, and $N_{ph}=3$ phases. This classifies the motor as having $N_{spp}=N_s/N_m/N_{ph}=1$ slot per pole per phase. Even

though not all slots are shown, the motor in Fig. 4-18 has slots distributed equally every 30°M around the stator as well, so it has $N_{spp}=1$ too. In Fig. 4-18, the winding is fractional pitch, but the slots are distributed just as they are in Fig. 4-12.

When N_{spp} is an integer, the motor is said to be an *integral slot motor*, and when N_{spp} has a fractional part, the motor is said to be a *fractional slot motor*. Note that there is a distinction between a motor having fractional pitch windings and a fractional slot motor. The first characterizes the windings; the second characterizes the slots that contain windings.

In an integral slot motor, the back EMFs of all coils making up a phase winding are in phase with each other, even if the winding itself is fractional as shown in Fig. 4-18. However, in a fractional slot motor, the back EMF of coils making up a phase winding are not all in phase with each other. Therefore, connecting all coils in series does *not* produce a back EMF that is simply equal to an amplitude-scaled version of the back EMF of one coil. In fractional slot motors, the net back EMF has a different shape as well as a different amplitude. It is this feature of fractional slot motors that lets the designer shape the back EMF.

To understand how a fractional slot motor differs from an integral slot motor, consider the motor shown in Fig. 4-21, which has $N_s=15$ slots, $N_m=4$ magnet poles, and $N_{ph}=3$ phases. For this motor, the number of slots per pole per phase is $N_{spp}=1.25$ and the angular slot pitch θ_s is $360^\circ\text{M}/N_s=24^\circ\text{M}$ or 48°E .

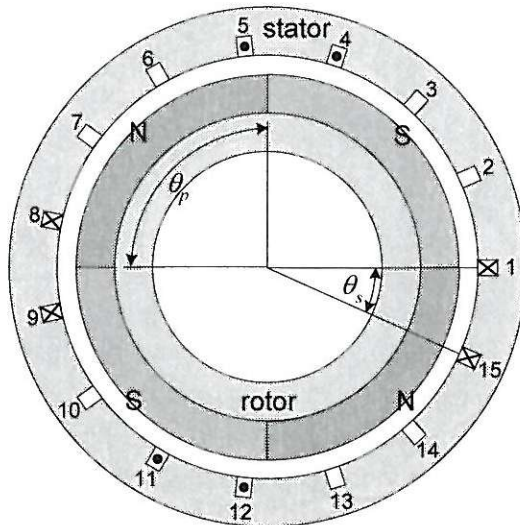


Figure 4-21. A fractional slot motor.

Because the slot pitch is not a integral subdivision of the angular pole pitch θ_p , it is not possible to use full pitch coils. In fact, by inspection it is not obvious how the motor should be wound. If a coil spans three slots, the angular coil pitch will be $3 \cdot 48^\circ\text{E}$ or 144°E . On the other hand, if a coil spans four slots, the coil pitch will be $4 \cdot 48^\circ\text{E}$ or 192°E . A three slot coil span is less than $\theta_p = 180^\circ\text{E}$, and a four slot coil span is greater than $\theta_p = 180^\circ\text{E}$.

Postponing winding layout design to a later chapter, Fig. 4-22 shows a valid winding layout for phase A of the motor. Since there are 3 phases, 15 slots, and each coil fills a net one slot (*i.e.*, each coil side fills one half slot), there are $N_{cph} = N_s / N_{ph} = 15/3 = 5$ coils per phase. Each coil in the figure spans three slots, so the coil pitch or throw is three slots.

Based on the procedure demonstrated previously, Fig. 4-23 shows the flux linkage and back EMF for the coil labeled C_a . The other coils have the same span as coil C_a , so their individual flux linkage and back EMFs have the same shape and amplitude, but are shifted in phase based on their angular position relative to coil C_a . As shown in Fig. 4-22, the center of coil C_b is offset 4 slots from the center of coil C_a , making $\theta_{ab} = 4 \cdot 48^\circ\text{E} = 192^\circ\text{E}$. Similarly, $\theta_{ac} = 7 \cdot 48^\circ\text{E} = 336^\circ\text{E}$, $\theta_{ad} = 8 \cdot 48^\circ\text{E} = 384^\circ\text{E}$, and $\theta_{ae} = 11 \cdot 48^\circ\text{E} = 528^\circ\text{E}$. Furthermore, coils C_b and C_e are wound in the opposite direction, so their back EMF has the opposite sign of that of coil C_a .

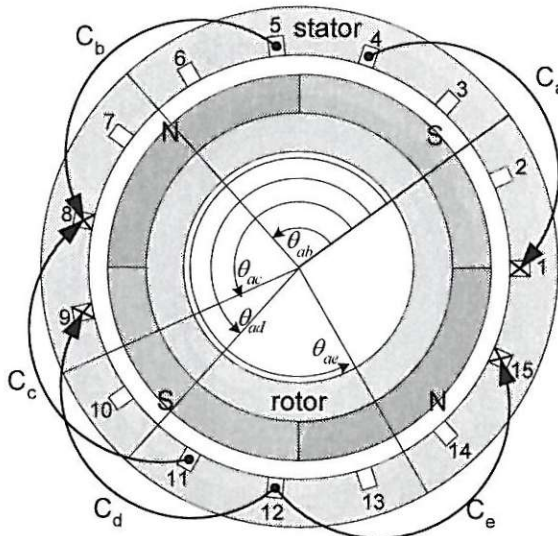


Figure 4-22. Phase A winding for a 4 pole, 15 slot motor.

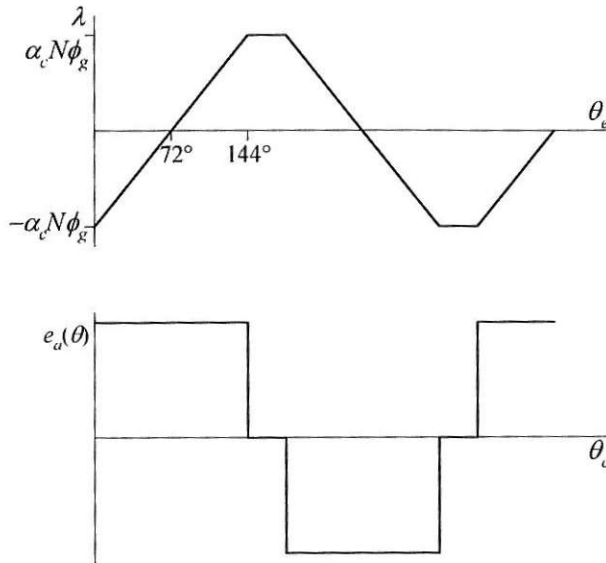


Figure 4-23. Flux linkage and back EMF of coil C_a .

Based on this information, if $e_a(\theta)$ is the back EMF shown in Fig. 4-23 for coil C_a , then the back EMF for the other coils are related to the back EMF of phase A by the phase shifts $-e_a(\theta-192^\circ)$, $e_a(\theta-336^\circ)$, $e_a(\theta-384^\circ)$, and $-e_a(\theta-528^\circ)$ for coils C_b , C_c , C_d , and C_e respectively. If the five coils are connected in series the net winding back EMF is given by the sum of the individual back EMFs,

$$\begin{aligned} e(\theta) &= e_a(\theta) - e_a(\theta - 192^\circ) + e_a(\theta - 336^\circ) + e_a(\theta - 384^\circ) - e_a(\theta - 528^\circ) \\ &= e_a(\theta) + e_a(\theta - 12^\circ) + e_a(\theta + 24^\circ) + e_a(\theta - 24^\circ) + e_a(\theta + 12^\circ) \end{aligned} \quad (4.12)$$

where the properties $-f(\theta) = f(\theta \pm 180^\circ)$ and $f(\theta) = f(\theta \pm 360^\circ)$ were used to simplify the first expression. This equation shows that the net winding back EMF is the sum of five phase A back EMFs respectively shifted by $0, \pm 12^\circ$, and $\pm 24^\circ$.

With the phase A back EMF shown in Fig. 4-23, Fig. 4-24 shows the net winding back EMF. The first five waveforms are the individual coil back EMFs, and the last is the net back EMF. The final result looks much more sinusoidal than the individual coil back EMFs. In fact, the net back EMF of an actual four pole, fifteen slot motor looks very sinusoidal due to the inherent leakage flux that is ignored in this analysis.

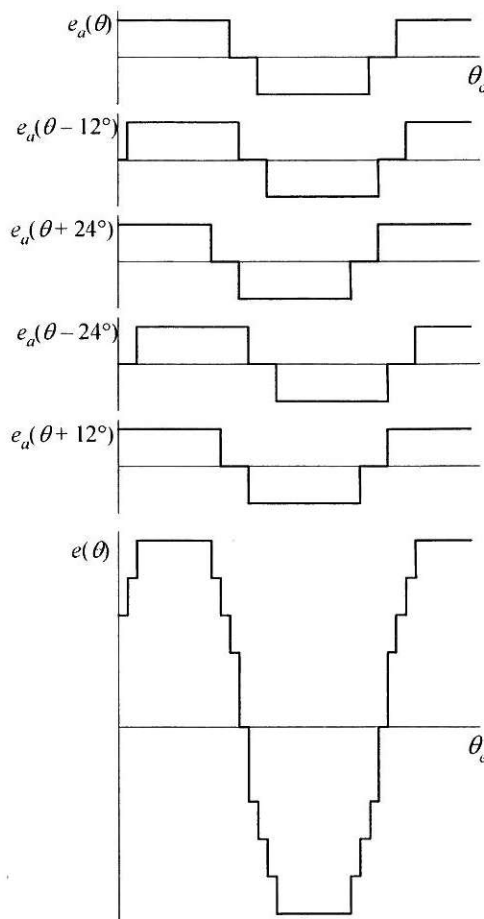


Figure 4-24. Sum of coil back EMFs to get net winding back EMF.

4.5 Coil Resistance

Multiple coils connected together to form phases are a basic part of all motors. Coils have two electrical properties, namely resistance and inductance. Of these, resistance is straightforward to describe, whereas inductance is harder to describe especially when mutual inductance is present as is common in motors.

Resistance is a property of all materials. It represents a measure of how much the material resists the flow of current. For conductive materials such as copper, the resistance R of a block of material in ohms (Ω) is given by

$$R = \frac{\rho l}{A} \quad (4.13)$$

where ρ is the resistivity of the material in $\Omega \cdot \text{m}$, l is the material length along the direction of current flow, and A is the cross-sectional area through which current flows. In general, material resistivity is a function of temperature, with resistivity increasing exponentially with increasing temperature. For copper and aluminum wire, resistivity versus temperature is approximated by the linear relationship

$$\rho(T) = \rho(T_0)[1 + \alpha(T - T_0)] \quad (4.14)$$

where T is temperature, T_0 is a base temperature, and α is the thermal resistivity coefficient. For copper, the parameters in this equation are commonly chosen as $T_0 = 20^\circ\text{C}$, $\rho(T_0) = 1.7241 \cdot 10^{-8} \Omega \cdot \text{m}$, and $\alpha = 4 \cdot 10^{-3}$. As material temperature increases over 100°C , this expression increasingly underestimates resistivity. Given α , copper wire resistance increases 4% for each 10°C ; therefore a wire at 120°C has 40% greater resistance than it does at 20°C .

On the basis of the above relationships, coils in motors are most commonly composed of multiple turns of round insulated wire as shown in Fig. 4-25. The center-most circle in the figure is the bare conductor having diameter d_{wb} and cross-sectional area A_{wb} . The next outer layer is the wire insulation, which is commonly available in three thicknesses, single, double (or heavy), and triple. The final outer layer is an optional layer of bonding material. This bonding layer is commonly composed of an adhesive that serves to bond layers of wire together after coils are formed. As shown in the figure, d_{wc} and A_{wc} are the covered wire diameter and cross-sectional area respectively, including the bonding layer.

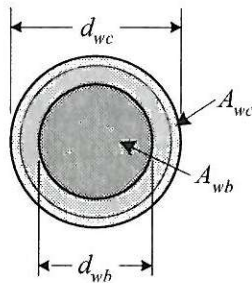


Figure 4-25. Wire cross section showing conductor, insulation, and bonding.

Several standards exist for classifying wire according to diameter. One common standard is American Wire Gage (AWG). In AWG, standard wire diameters form a geometric progression described by the relationship

$$d_{wb} = 8.251463(0.8905257)^G \quad (4.15)$$

where G is the integer wire gage, and d_{wb} is the bare wire diameter in mm. There is an inverse relationship between wire gage and diameter. As the gage increases, the diameter decreases. Alternatively, as the gage increases, the resistance per unit length increases. The inverse of the above relationship is

$$G = \frac{\log\left(\frac{d_{wb}}{8.251463}\right)}{\log(0.8905257)} \quad (4.16)$$

Because AWG is based on a geometric progression, wire gages are related to each other by ratios as shown in Fig. 4-26. This figure plots resistance relative to a wire having any gage G to wires having gages G , $G+1$, etc. The most notable point on the curve appears at $G+3$. A wire of gage $G+3$ has twice the resistance of a wire of gage G . So two wires of gage $G+3$ taken in parallel have the same resistance as one wire of gage G . Other important points appear at $G-1$ and $G+1$. At $G+1$, resistance is approximately 26% greater than that at G . So increasing the wire gage by one, increases resistance and I^2R losses by 26% provided current remains constant. Alternatively, at $G-1$, resistance decreases approximately to about 79% of that at G . Therefore, decreasing the wire gage by one, decreases the I^2R losses for fixed current to 79% of what they are at G .

The current capacity of the wire depends on its cross-sectional area and its thermal environment. The heat density in a resistance, *i.e.*, I^2R loss per unit volume, is equal to ρJ^2 , where J is the current density in the material. Based on experience, the maximum allowable current density varies roughly between 1 and 10 Arms/mm². Using these limits as a guideline, Fig. 4-27 shows the allowable RMS wire current versus wire gage. In confined volumes with little thermal conductivity, the lower limit of 1 Arms/mm² may be too high. Similarly, if wire is actively cooled, 10 Arms/mm² may be overly conservative as an upper limit. It is interesting to note that the rated current for 14 gage household wiring is 15 Arms, which corresponds to a current density of 7.2 Arms/mm².

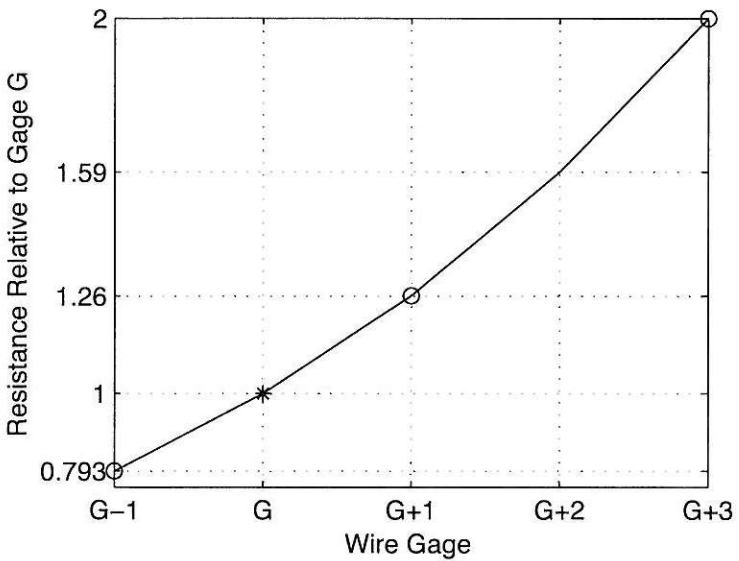


Figure 4-26. Relative wire resistance versus wire gage.

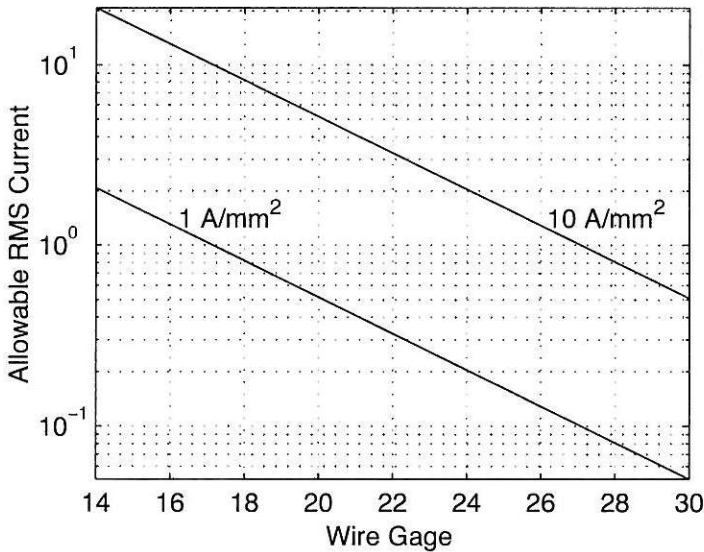


Figure 4-27. Current capacity versus wire gage.

In motor design, computing coil resistance is simply a matter of applying (4.13) through (4.16) and taking into account the added diameter of the insulation and optional bonding material. Since the insulation and bonding thicknesses do not fol-

low a geometric progression, (4.15) or a similar equation cannot be used to compute the covered diameter d_{wc} . The single, double, and triple insulation thicknesses and the bonding thicknesses do not conform to any smooth function with respect to wire gage, so they are best determined by consulting a manufacturer's wire table.

4.6 Coil Inductance

Inductance is not usually a critical parameter in brushless permanent magnet motors. Inductance determines the time constant of the windings, and therefore, determines the rate at which winding currents can change. Modern power electronic control techniques that rely on current control principles are relatively insensitive to the exact inductance value. As a result, inductance calculations are often simplified to produce analytical inductance estimates. If greater accuracy is required, three-dimensional finite element analysis is required.

When a coil is placed in stator slots, its inductance changes dramatically compared to its inductance when surrounded by air. Computing coil inductance is a matter of determining the magnetic field created by coil current and then relating this field to inductance through the use of (3.4). Alternatively, inductance can be computed by equating the coenergy stored in an inductance (3.26) to the coenergy stored in the field created by the coil (3.27). That is, inductance can be found by solving

$$\frac{1}{2} Li^2 = \int_V \frac{1}{2} \mu H^2 dV \quad (4.17)$$

for the inductance L , where V is the volume containing the field.

In addition to self inductance as described above, mutual inductance exists between the coils in a given phase as well as between the coils in different phases. Mutual inductance between coils in a given phase is considered here, but mutual inductance between coils in different phases is not. Because mutual inductance between phases is small relative to self inductance (*e.g.*, around 10%) and because mutual inductance requires much more careful computation using these same general procedures, it will not be considered here.

When coils are placed in slots, the coil inductance has three distinct components due to the three distinct areas where significant magnetic field is created by coil current. These three areas are the air gap, the slots, and the end turns. The ferromagnetic portions of the motor do not contribute to the inductance as long as their relative permeability is high. Moreover, magnet flux does not contribute to inductance because inductance relates coil current to the flux linkage created by the coil current acting

alone. From the coil's perspective, the magnets are simply blocks of material having relative permeability of $\mu_R\mu_0$.

Air Gap Inductance

To illustrate the computation of inductance, consider the 4 pole, 12 slot motor having full pitch windings as shown in Fig. 4-11. The air gap inductance component is due to the flux crossing the air gap. As a result, the air gap flux must be found using a magnetic circuit model. This circuit must model the MMF produced by the stator coils and ignore the flux source in the magnet model. Figure 4-28 shows the associated magnet circuit model, where R_g is the air gap reluctance over one pole pitch, R_m is the magnet reluctance, and Ni is the MMF source associated with each coil. The polarity of each MMF source corresponds to the application of the right hand rule to each coil.

Letting the outer ring be the reference node, identifying F_r as the MMF at the center node, and setting the sum of the fluxes leaving the center node to zero (*i.e.*, Kirchhoff's current law applied to flux) gives

$$\phi_1 - \phi_2 + \phi_3 - \phi_4 = 0$$

$$\frac{F_r + Ni}{R_g + R_m} - \frac{Ni - F_r}{R_g + R_m} + \frac{F_r + Ni}{R_g + R_m} - \frac{Ni - F_r}{R_g + R_m} = 0 \quad (4.18)$$

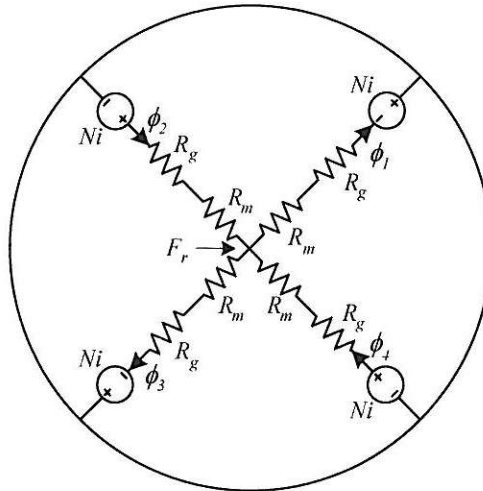


Figure 4-28. Magnetic circuit model for the computation of air gap inductance.

Simplifying the last expression leads to the result $F_r=0$, which implies that

$$\phi_1 = \phi_2 = \phi_3 = \phi_4 = \frac{Ni}{R_g + R_m} \quad (4.19)$$

Therefore, the net flux linked by all four coils is

$$\lambda = N(\phi_1 + \phi_2 + \phi_3 + \phi_4) = \frac{N_m N^2 i}{R_g + R_m} \quad (4.20)$$

where $N_m=4$ is the number of coils (and magnet poles). Using the definition of inductance (3.4), the *air gap inductance* is

$$L_g = \frac{\lambda}{i} = \frac{N_m N^2}{R_g + R_m} \quad (4.21)$$

Using expressions for the magnet and air gap reluctances (4.2) and the air gap cross-sectional area $A_g = L_{st} \theta_p R_{ro} = (2\pi/N_m) L_{st} R_{ro}$, where L_{st} is the axial motor length and R_{ro} is the air gap radius as shown in Fig. 4-6, yields

$$L_g = \frac{2\pi\mu_0 L_{st} R_{ro}}{g + \frac{l_m}{\mu_R C_\phi}} N^2 \quad (4.22)$$

where N is the number of turns per coil. This air gap inductance assumes that all four coils are connected in series and is therefore the air gap inductance of the complete phase winding. This expression applies to all full pitch winding cases since N_m does not appear in (4.22). In addition, the air gap inductance is proportional to the rotor surface area, since $2\pi R_{ro} L_{st}$ appears in the numerator. The denominator is the magnetic length from the stator surface through the air gap and magnet to the rotor yoke.

Modifications to the above equation are required when the individual coils are connected in parallel or in some combination of series and parallel. These modifications are covered later in this chapter.

Slot Leakage Inductance

In addition to the air gap, coil current produces a magnetic field that crosses from one side of a slot to the other as shown in Fig. 4-29. This field makes sense when one considers that a magnetic field circulates around conductors, which in this case are

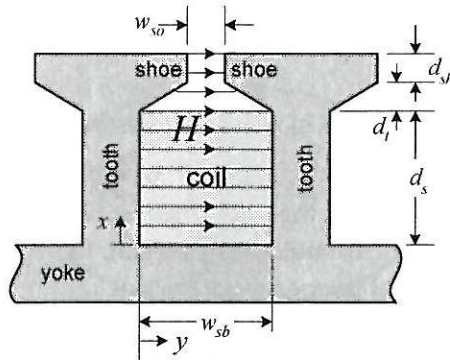


Figure 4-29. Slot leakage flux.

carrying current into the page. The figure does not depict the magnetic field in the teeth and stator yoke because the field intensity H is near zero as a result of the high relative permeability assumed for the ferromagnetic portions of the motor. For simplicity, this figure depicts a slot in a linear motor. The analysis that follows applies to rotational motors, with the associated coordinate system changes. This figure includes narrow slot openings between shoes that taper back to the stator teeth.

The inductance component that results from the magnetic field that crosses the slot in the y -direction is commonly called the *slot leakage inductance*. Computation of this component requires more work than calculating the air gap inductance because the magnetic field linking the coil is not constant over the cross section of the coil, rather it is approximately linear with respect to slot depth, which can be described as

$$H_y(x) = \frac{(2N)i}{w_{sb}} \frac{x}{d_s} \quad (4.23)$$

where the slot is assumed to contain two coil sides each having N turns. As stated, the field intensity H_y is zero at the slot bottom because no current is enclosed. As x increases, more current is enclosed. Finally, when all the current is enclosed at $x=d_s$, the field intensity reaches its maximum value equal of $H_y=Ni/w_{sb}$. In the shoe area, the field intensity has a constant amplitude equal to this maximum value.

Computing the inductance in this coil area of the slot requires using the coenergy relationship (4.17). In this case, the volume integral is the slot volume occupied by the coil. Performing the required integration and solving for the inductance gives the coil area leakage inductance of

$$L_{ca} = \frac{\mu_0 d_s L_{st}}{3w_{sb}} (2N)^2 \quad (4.24)$$

where L_{st} is the axial length of the slot. It is interesting to note that this expression matches the fundamental relationship $L_{ca} = N^2 P$ where P in this case is the effective permeance of the slot, often called the *slot permeance coefficient*. Because the winding is assumed to be uniformly distributed over the slot, the effective permeance is one third of the standard or normal permeance of the slot.

Because the field intensity is constant over the shoe area, the inductance of this area can be approximated using the fundamental $N^2 P$ relationship. Doing so and including the coil area term leads to a total per-slot slot leakage inductance of

$$L_{s1} = (2N)^2 \left[\frac{\mu_0 d_s L_{st}}{3w_{sb}} + \frac{\mu_0 d_t L_{st}}{(w_{so} + w_{sb})/2} + \frac{\mu_0 d_{sh} L_{st}}{w_{so}} \right] \quad (4.25)$$

where the first term inside the brackets is the effective permeance of the coil area, the second term is an approximate permeance for the tapered area, and the last term is the permeance of the shoe tip area. For the full pitch winding being considered, there are N_m slots each having the slot leakage inductance (4.25). Taking this into account gives the slot leakage inductance of the winding as

$$L_s = N_m (2N)^2 \left[\frac{\mu_0 d_s L_{st}}{3w_{sb}} + \frac{\mu_0 d_t L_{st}}{(w_{so} + w_{sb})/2} + \frac{\mu_0 d_{sh} L_{st}}{w_{so}} \right] \quad (4.26)$$

End Turn Inductance

The end turn inductance is the only remaining inductance component. This inductance is created by the magnetic field that surrounds a coil after it leaves one slot and before it enters another slot. Because the end turn layout is subject to few restrictions, a set magnetic field distribution is difficult to define. As a result, the end turn inductance is often a rough approximation. The approach followed here is to use the coenergy relationship (4.17) and to assume that the magnetic field is distributed in the same way as it is about an infinitely long cylinder having a surface current i , as illustrated in Fig. 4-30. For this configuration, the circumferential field intensity is

$$H_\theta = \frac{i}{2\pi r} \quad (4.27)$$

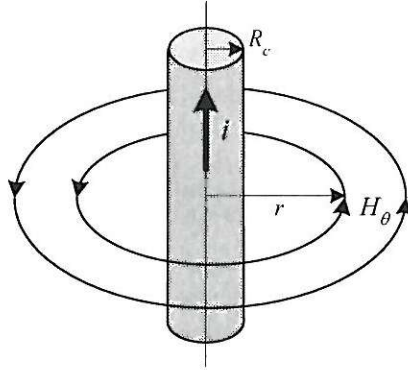


Figure 4-30. Magnetic field about a cylindrical conductor.

for r greater than the radius of the conductor R_c . For a conductor of length l_c , a radius $r > R_c$, and total current equal to Ni , the inductance is

$$L = \frac{\mu_0 l_c N^2}{2\pi} \ln \left(\frac{r}{R_c} \right) \quad (4.28)$$

Consider the geometry shown in Fig. 4-31. The end turns are semicircular with radius equal one half the mean coil pitch τ_{cp} ; the radius r is equal to $\tau_{cp}/2$; and because there are two coils per slot, the end turn bundle has a cross-sectional area equal to one half the slot cross-sectional area A_s . The end turn inductance for one end turn bundle having N turns is

$$L_{e1} = \frac{\mu_0 \tau_{cp} N^2}{4} \ln \left(\frac{\tau_{cp} \sqrt{\pi}}{\sqrt{2A_s}} \right) \quad (4.29)$$

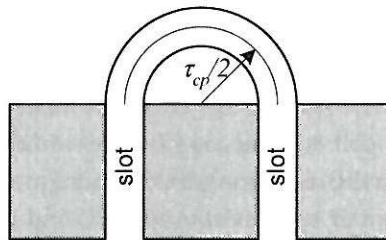


Figure 4-31. End turn geometry approximation.

Since there are $2N_m$ end turn bundles per phase winding and there is no mutual coupling between the end turns of other coils in the same phase, the total end turn inductance per phase is

$$L_e = \frac{N_m \mu_0 \tau_{cp} N^2}{2} \ln \left(\frac{\tau_{cp} \sqrt{\pi}}{\sqrt{2} A_s} \right) \quad (4.30)$$

Given (4.22), (4.26) and (4.30), the net phase winding inductance with all coils connected in series is the sum of the three fundamental components,

$$L_{ph} = L_g + L_s + L_e \quad (4.31)$$

4.7 Series and Parallel Connections

In the preceding sections, back EMF, resistance and inductance were computed under the assumption that all coils in a phase are connected in series. This arrangement occurs in the majority of motor designs. Occasionally, the coils are connected in other ways. For example, if there are four coils per phase, they can be connected in three different ways: all coils in series, all coils in parallel, and combinations of two coils in series connected in parallel with the other two coils connected in series. Of these combinations, by far the most common is all coils in series, followed by all coils in parallel. Only in exceptional circumstances are coils connected in combinations of series and parallel.

The overwhelming choice of all coils in series is due to the interaction of the individual coil back EMFs. When all coils are connected in series, the phase back EMF is simply the sum of the individual coil back EMFs. For coils having identical phase relationships such as in Fig. 4-11, this means scaling the amplitude of the coil back EMF by the number of coils. For coils having different phase relationships, this addition changes the shape of the back EMF as well, as illustrated in Fig. 4-24.

When coils are connected in parallel, the coil back EMFs can create circulating currents that contribute to $I^2 R$ losses but do not provide beneficial torque production. To understand how this occurs, consider the circuit model for two coils in parallel as shown in Fig. 4-32a. Each coil has an associated resistance, inductance, and back EMF. By ignoring the external connections, the circuit simplifies to that shown in Fig. 4-32b. The two inductances and two resistances add, and the two back EMF sources subtract. When the two coil back EMFs have the same amplitude and phase relationships, *i.e.*, they are instantaneously identical, the combined back EMF $e_1 - e_2$ is equal to

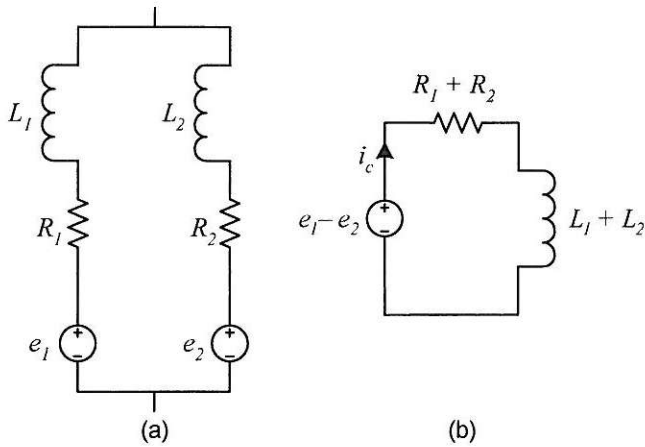


Figure 4-32. Two coils connected in parallel.

zero and no current i_c circulates around the loop from one coil to the other. However, if the individual coil back EMFs are not instantaneously identical, the combined back EMF $e_1 - e_2$ is nonzero and current i_c circulates around the loop independent of any current applied to the parallel coils during motor operation.

For example, if the back EMFs $e_a(\theta - 24^\circ)$ and $e_a(\theta + 24^\circ)$ for the four pole, fifteen slot motor shown in Fig. 4-24 are connected in parallel, their difference is nonzero as shown in Fig. 4-33. This back EMF difference creates a circulating current that flows through the resistance $R_1 + R_2$, creating an undesirable I^2R loss.

To avoid circulating currents and their associated loss, only coils having identical back EMFs can be connected in parallel. For most motor designs this is not possible and therefore parallel-connected coils do not appear often in practice. In the unusual case when the number of turns cannot be decreased to lower the back EMF amplitude, parallel coil connections must be accepted. In this case, the undesirable I^2R loss must be accepted or the motor design must be restricted so that the back EMF of individual coils are identical.

In the situation where all coils in a phase are connected in parallel, the resistance and inductances are modified from their previously computed expressions which applied to all coils connected in series. The resistance relationship between all coils in series versus all coils in parallel can be determined from Fig. 4-34. The resistors on the left represent n individual coil resistances connected in series, whereas the resistors on the right represent the n individual coil resistances connected in parallel. When the coils are identical, all the coil resistances have identical values. If R_{eqs} is the

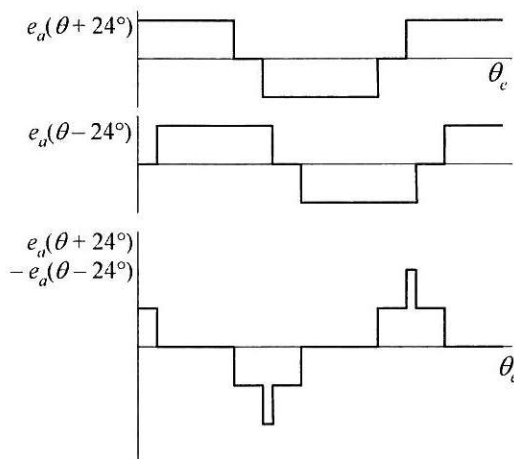


Figure 4-33. An example of coil back EMFs that lead to circulating currents.

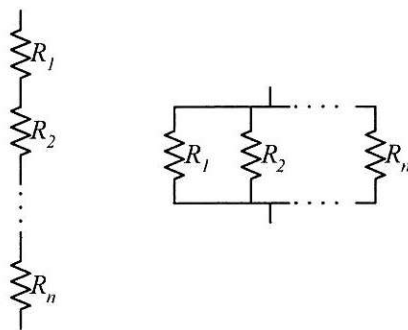


Figure 4-34. Coil resistances in series and in parallel.

equivalent resistance of all coils connected in series and R_{eqp} is the equivalent resistance of all coils connected in parallel, then R_{eqp} is related to R_{eqs} by

$$\frac{R_{eqp}}{R_{eqs}} = \frac{R/n}{nR} = \frac{1}{n^2} \quad (4.32)$$

where R is the resistance of each coil. Therefore, conversion of phase resistance computed as all coils in series to all coils in parallel requires division by the square of the number of coils, or *number of parallel paths*.

Since inductances add in series and parallel in the same way that resistances do, conversion of phase inductance computed with all coils in series to all coils in paral-

lel should follow the same relationship given in (4.32). Indeed, this is true if all the coil inductances are independent of one another; that is, they do not share common flux. However (4.32) does not apply if the coils share common flux, as is the case for the air gap and slot components of the phase inductance.

Rigorous computation of the phase inductance for coils connected in parallel requires significant effort to keep track of components that share mutual flux and those that do not. Given the infrequency of connecting coils in parallel, the fact that inductance is a less critical aspect of motor design, and that some inductance components require significant approximations for their computation, it is common to forego the needed rigor and simply use (4.32) to approximate the phase inductance of all coils connected in series to all coils connected in parallel. This mutual flux issue becomes even more complicated when combinations of series and parallel connections among coils are implemented in a motor design.

As a final thought on coil connections, it is interesting to note that connecting coils in series or in parallel has no influence on the electrical time constant $\tau = L/R$ of a phase winding. That is, assuming (4.32) holds for both phase resistance and phase inductance, the time constant of the phase winding does not change no matter how the coils are connected to form a phase winding $L_{eqs}/R_{eqs} = L_{eqp}/R_{eqp}$. As a result, there is no incentive to connect coils in a way that differs from all coils in series to achieve a different time constant.

4.8 Armature Reaction

Armature reaction refers to the magnetic field produced by ampere-turns Ni in the stator slots and its interaction with the permanent magnet field. For example, the air gap flux due to coil current was illustrated earlier in the section on air gap inductance. The flux that crosses the air gap due to coil current in the four pole, twelve slot motor is given by (4.19). Dividing this expression by the cross-sectional area of the magnet surface through which this flux flows gives the armature reaction flux density in the air gap. Because the magnet recoil permeability is near one and the magnet length is larger than the air gap length, the armature reaction flux density in the air gap of a motor with surface-mounted magnets is usually small under normal operating conditions, *e.g.*, usually less than ten percent of the air gap flux density due to the permanent magnet.

In addition to the air gap flux created, coil current produces additional flux in the stator shoes and teeth. Once again, in common motor designs, this flux is significantly lower than the shoe and tooth flux coming from the rotor magnets.

Ideally, the magnetic field distribution within the motor is the linear superposition of the magnetic fields due to the permanent magnets and the coil currents. In reality, the presence of saturating ferromagnetic material in the stator can cause these two fields to interact nonlinearly. When this occurs, the performance of the motor diminishes.

Generally speaking, as long as the stator teeth and shoes are not highly saturated due to the permanent magnets acting alone, armature reaction is not a problem. The greatest concern with respect to armature reaction normally occurs under fault conditions. If the power electronics used to drive the motor enters a fault condition where the motor currents exceed their normal range by an order of magnitude or greater, the armature reaction magnetic field can become large enough to demagnetize the rotor magnets.

4.9 Slot Constraints

Slot Fill Factors

The use of slots to hold coils places fundamental constraints on motor design. These constraints are best understood by considering the representative slot shown in Fig. 4-35. It is readily apparent from the figure that round wire cannot fill the slot entirely. Gaps exist between wire turns and the nonuniform slot shape limits the ability to place turns uniformly. As a result, it is convenient to relate the slot cross-sectional area to the wire cross-sectional area by defining a *bare wire slot fill factor*

$$K_{wb} = \frac{NA_{wb}}{A_s} \quad (4.33)$$

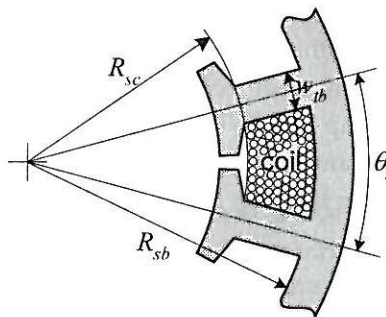


Figure 4-35. A representative slot.

where A_s is the slot cross-sectional area, N is the number of turns in the slot, and A_{wb} is the bare wire cross-sectional area. In addition, it is convenient to define a *covered wire slot fill factor*

$$K_{wc} = \frac{NA_{wc}}{A_s} \quad (4.34)$$

where A_{wc} is the covered wire cross-sectional area. Clearly, wire insulation takes up nonzero space, $K_{wb} < K_{wc}$. For rectangular slots uniformly filled with wire stacked in rows and columns, there is an upper limit to K_{wc} . As shown in Fig. 4-36, the upper limit is given by the ratio of the covered wire cross-sectional area to that of the square area enclosing it. That is,

$$K_{wc-\max} = \frac{\pi(d_{wc}/2)^2}{d_{wc}^2} = \frac{\pi}{4} \approx 79\% \quad (4.35)$$

In practice, the achievable slot fill factors are dependent on the technology used to wind coils within slots or insert preformed coils into slots and the price one is willing to pay. The achievable slot fill is also decreased by the presence of *slot liners*, i.e., the addition of insulating material to the inner slot surface to eliminate abrasion and electrical shorts between motor windings and the stator laminations. Without incurring extreme cost, it is generally possible to achieve in the neighborhood of $K_{wb} = 50\%$ and $K_{wc} = 60\%$, depending on the motor size, slot shape and number of turns.

Slot Resistance

Based on the above relationships and the resistance relationship (4.13), the total resistance in a slot of length L_{st} containing N turns connected in series is

$$R_{slot} = \frac{\rho N L_{st}}{A_{wb}} = \frac{\rho L_{st} N^2}{K_{wb} A_s} \quad (4.36)$$

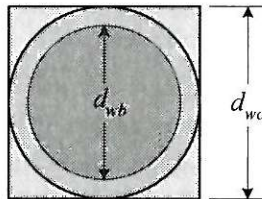


Figure 4-36. Covered wire and the square enclosing it.

This equation shows that coil resistance is proportional to the number of turns squared. Since all components of coil inductance are proportional to N^2 as well, the electrical time constant $\tau=L/R$ of windings is independent of the number of turns, provided the slot fill and slot cross-sectional area remain fixed. For example, a coil formed from N turns of gage G wire and a coil formed from $2N$ turns of gage $G+3$ wire (*i.e.*, wire having one half the cross-sectional area) will have the same time constant since the resistance and inductance both increase by a factor of four.

The power dissipated as heat by the coil resistance, *i.e.*, the I^2R losses, are also independent of the number of turns N . Using (4.36), the I^2R losses per slot are given by

$$P_{slot} = I^2 R_{slot} = I^2 \frac{\rho N L_{st}}{A_{wb}} \quad (4.37)$$

where I is the RMS current in the wire. By defining the RMS conductor current density as $J=I/A_{wb}$ and identifying the bare copper volume in the slot as $V_{wb}=N L_{st} A_{wb}$, (4.37) can be written as

$$P_{slot} = \rho V_{wb} J^2 \quad (4.38)$$

This equation confirms that the I^2R losses in a winding are independent of N , and therefore, there is no way to manipulate N to improve motor efficiency. Moreover, dividing both sides of (4.38) by V_{wb} , gives a slot heat density of ρJ^2 in W/m^3 which was cited earlier in the discussion of wire capacity versus wire gage as shown in Fig. 4-27.

Wire Gage Relationships

Given a constant slot fill factor, a fixed slot cross-sectional area, and the wire gage relationships (4.15) and (4.16), it is possible to relate the relative number of turns that fill a slot to relative wire gage. The number of turns that fit in a slot is inversely proportional to wire cross-sectional area or the wire gage. Figure 4-37 depicts this relationship. For example, if N turns of gage G fill a slot, then approximately 26% more turns of wire gage $G+1$ fill the slot equally well.

In addition, it is possible to relate relative coil resistance to relative wire gage. This relationship is shown in Fig. 4-38. For example, if the coil resistance using wire of gage G is R , then increasing the wire gage to $G+1$, increases the resistance by about 59% to approximately $1.59R$.

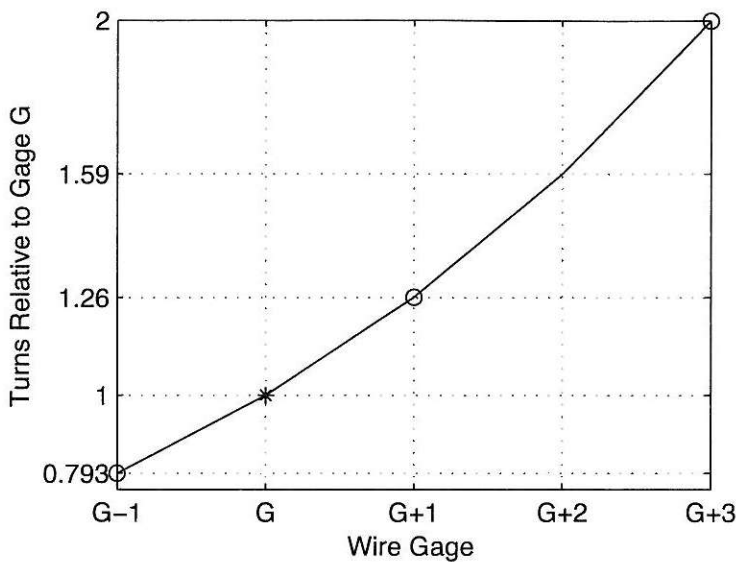


Figure 4-37. Relative number of turns required to fill a slot equally well.

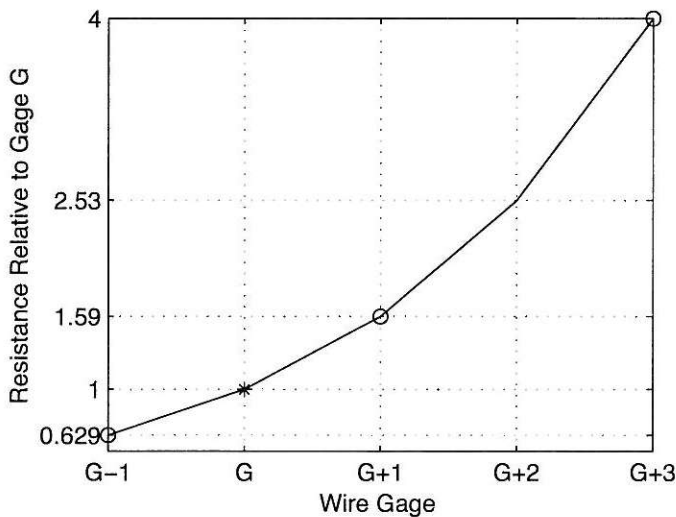


Figure 4-38. Relative slot resistance for a fixed slot area and fill factor.

Constancy of Ni

A further constraint imposed by fixed slot cross-sectional area is the constancy or invariance of the product Ni . This product is the MMF produced by a coil having N

turns, each carrying a current of i . Based on the bare wire slot fill factor (4.33) and the current density relationship, the product Ni is equal to

$$Ni = NA_{wb}J = K_{wb}A_sJ \quad (4.39)$$

This equation shows that Ni is a constant set by the bare wire slot fill factor and the slot current density. When N increases by some factor, the wire cross-sectional area and resulting current decrease by the same factor. The constancy of Ni implies that neither N nor i can be changed independently to maximize torque as given by (4.8).

4.10 Torque Constant, Back EMF Constant, and Motor Constant

The specification sheets for brushless permanent magnet motors often cite values for the torque constant $K_t = T/i$ and the back EMF constant $K_e = E_b/\omega_m$, but few provide values for the motor constant K_m . Of these specifications, the torque constant is generally the least useful and most deceiving. At first glance it appears that a larger torque constant means that more torque can be obtained for the same amount of current. However, for a given rotor and stator, the constancy of Ni implies that this is not true. Increasing the torque constant by increasing N does not increase the torque capability of a motor since any increase in N forces the current i to decrease correspondingly.

In the selection of a motor, the back EMF constant is generally more useful because it does not involve the current i . For a given rotor and stator, the back EMF constant scales linearly with N . As a result, by winding the motor with different gage wires and keeping the slot fill factor constant, smaller diameter wires make room for greater numbers of turns N . As a result, different back EMF constants are obtained. Therefore, in practice, the back EMF constant and rated application speed are used to set the peak back EMF voltage to some level just under the voltage available from the power electronics used to drive the motor. Once the back EMF constant is determined by the power electronics, the torque constant is set because of their equivalence as given in (4.9). (Note that motor specification sheets do not demonstrate this equivalence because the torque constant and back EMF constants are specified using inconsistent conditions and inconsistent units.)

Since the torque constant and back EMF constant vary with N and the product Ni is constant, they are not good indicators of motor performance. A more useful performance indicator is *motor constant*, which is defined as

$$K_m = \frac{T}{\sqrt{I^2 R}} \quad (4.40)$$

As stated, the motor constant expresses torque efficiency, which is different from power efficiency. Motor constant describes how efficiently a motor produces torque as a function of the I^2R losses incurred in the production of that torque. That is, motor constant is related not only to the desired motor output, but is also inversely proportional to the cost of producing that output.

Since torque is proportional to current and power is proportional to current squared, using the square root of power in (4.40) allows current to cancel from the ratio. As a result, motor constant is independent of current and the number of turns N , and remains unchanged as long as the slot fill factor is fixed. Therefore, motor constant provides a good indication of motor performance.

To illustrate these facts, consider the ideal torque produced by one coil filling two slots as given in (4.8) and having twice the total resistance of one slot (4.36). Substituting these expressions into (4.40) and simplifying gives

$$K_m = \frac{2NB_g L_{st} R_{ro} I}{\sqrt{I^2 (2R_{slot})}} = \frac{2NB_g L_{st} R_{ro}}{\sqrt{2\rho L_{st} N / A_{wb}}} = \frac{B_g R_{ro}}{\sqrt{\rho}} \sqrt{V_{wb}} \quad (4.41)$$

where V_{wb} is the bare wire volume contained in the two slots. This result confirms that the motor constant is independent of number of turns N as well as current. A family of motors constructed from a given rotor and stator will produce torque with equal efficiency as long as the total volume of bare wire remains unchanged. As N increases, the torque per ampere (*i.e.*, torque constant) increases, but simultaneously the resistance increases with the square of N , so the motor constant is unchanged.

Torque efficiency as described by (4.41) is proportional to the air gap flux density, the rotor outside radius, and the square root of the wire volume. The air gap flux density can be increased by decreasing the air gap length, increasing the magnet length, (both of which imply increasing the permeance coefficient) or by using a more powerful magnet. The air gap length can only be decreased so far before manufacturing tolerances make it impossible to maintain a consistent air gap. Therefore, once the air gap is minimized, one must buy more magnet material or buy more powerful magnet material to increase torque efficiency.

Increasing the rotor outside radius increases the motor constant but requires that the stator diameter increase to maintain the wire volume. If the stator diameter is constant, increasing the rotor outside radius has little affect on the motor constant. If the rotor outside radius is fixed, increasing the wire volume means increasing the outside stator diameter or increasing the slot fill factor. Increasing the slot fill factor

alone is usually difficult because it becomes increasingly expensive to pack more and more wire into slots, especially as one gets closer to the theoretical maximum.

4.11 Torque per Unit Rotor Volume

In Chapter 1, it was argued that torque is proportional to rotor volume as stated in (1.8) as $T=kD^2L$. In this expression k is the only unknown. For a given rotor volume, its value determines the quality of a motor design. Rather than try to determine k specifically, it is beneficial to determine the torque produced per unit rotor volume. To do so, consider the torque produced by the motor shown in Fig. 4-11, whose phase torque is given by (4.11). Based on (4.11) the torque per unit rotor volume is

$$K_{TRV} = \frac{|T|}{\pi R_{ro}^2 L_{st}} = \frac{2N_m B_g Ni}{\pi R_{ro}} \quad (4.42)$$

This expression is directly proportional to k in (1.8). For a fixed air gap radius R_{ro} , the torque produced per unit rotor volume is directly proportional to the air gap flux density B_g , the number of magnet poles N_m , and the coil ampere-turns Ni . If the stator volume is fixed, Ni is a constant, thereby making N_m and B_g responsible for maximizing the torque per unit rotor volume. As with the motor constant, the air gap flux density plays a crucial role in determining performance. Increasing B_g means decreasing the air gap length, or increasing the quality or quantity of magnet material used.

The torque per unit rotor volume describes the amount of torque available from a given rotor volume. As such, it is a common measure for comparing motors and for initial motor sizing for a given application. Even more common is the use of the air gap shear stress, which is the tangential force per unit rotor surface area. This measure is often easier to visualize and is very simply related to the torque per unit rotor volume. Using (1.1) and (4.11), the air gap shear stress is given by

$$\sigma = \frac{F}{2\pi R_{ro} L_{st}} = \frac{N_m B_g NI}{\pi R_{ro}} = \frac{K_{TRV}}{2} \quad (4.43)$$

While this expression applies to the ideal conditions implied by (4.11), the air gap shear stress provides a useful comparative measure. Often the air gap shear stress is expressed in lbf/in^2 (psi). When expressed in these units, low cost brushless perma-

nent magnet motors typically exhibit a shear stress in the range $0.5 \leq \sigma \leq 2$, higher cost motors are in the range $1.5 \leq \sigma \leq 3$, very high performance motors are typically in the range $2 \leq \sigma \leq 10$, and large liquid cooled motors can be in the range $10 \leq \sigma \leq 20$. From (4.43) it is clear that the number of magnet poles, the air gap flux density, and the coil ampere turns Ni are important quantities. In low performance motors, these values are compromised to reduce cost, whereas in high performance motors, they are pushed to achieve performance.

4.12 Cogging Torque

As described in Chapter 1 and illustrated in Figs. 1-6 and 1-7, cogging torque describes the interaction of the rotor magnets acting on the stator teeth or poles independent of any current. Mathematically this torque was described as part of the general torque expression (3.37). While this torque is often considered beneficial in step motors, it is considered detrimental in brushless permanent magnet motors.

This dissatisfaction with cogging torque often lacks quantitative support. One of the first things engineers invariably do when they pick up a small motor is to spin the shaft with their fingers. The pulsations felt during this process are caused by cogging torque. In comparing several motors based on this qualitative examination, engineers will judge the one with the least cogging torque the best, even if it performs the worst in the actual motor application.

In reality, cogging torque is often very small relative to the beneficial mutual torque produced by a motor. Furthermore, even a slight mismatch between the back EMF of the motor and the motor current often produces greater ripple torque than the cogging torque itself. In this situation, cogging torque is masked by the larger torque variation in the mutual torque. As a result, the qualitative shaft spin test is often misleading.

Despite the insignificance of cogging torque in many applications and in the presence of mutual torque ripple due to back EMF–current mismatch, cogging torque is a motor characteristic worth understanding. Simply put, cogging torque is the torque created when the rotor permanent magnets attempt to align themselves with a maximum amount of ferromagnetic material. This is visually obvious for simple structures such as that shown in Fig. 1-6, but is more difficult to visualize for common motor structures despite the fact that the same fundamental phenomenon is occurring.

To understand cogging torque, consider the cogging torque term from (3.37)

$$T_{cog} = -\frac{1}{2}\phi^2 \frac{dR}{d\theta} \quad (4.44)$$

where ϕ is the magnet flux crossing the air gap and R is the total reluctance through which the flux passes. Clearly, if the reluctance R does not vary as the rotor rotates, the derivative in (4.44) is zero and the cogging torque is zero. In addition, cogging torque is independent of flux direction because the magnet flux ϕ is squared.

To illustrate how (4.44) applies to more common motor structures, consider the four pole, twelve slot motor considered earlier and as shown in Fig. 4-39. As each magnet in the motor rotates past the stator teeth, the reluctance experienced by the magnet under the slot openings changes because of the longer flux path length into the slots terminating on the shoes. Therefore, the slot openings create a varying reluctance for the magnet flux, thereby creating cogging torque. If the stator teeth did not have shoes, such as the motor shown in Fig. 1-11, the reluctance variation and resulting cogging torque would be much greater. Thus, the primary purpose for shoes is cogging torque reduction.

Shoe design represents a fundamental tradeoff. The narrower the slot opening w_{so} , the smaller the cogging torque becomes. In the limiting case, if there was no slot openings, cogging torque would be zero. On the other hand, the slot opening must

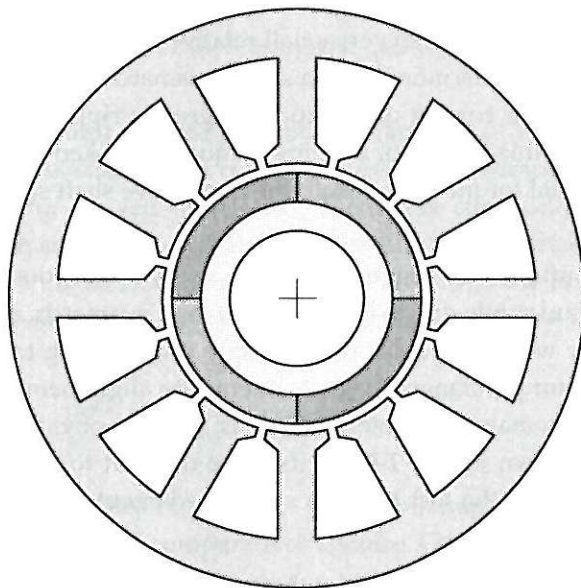


Figure 4-39. Cross section of a four pole, twelve slot motor.

be wide enough to insert coils, with the cost of inserting coils being directly proportional to the slot opening width. As a rule of thumb, the slot opening must be at least two to three times the covered diameter of the wire placed in the slots. A side effect of making the slot opening narrow to minimize cogging torque is that doing so increases the slot leakage inductance component of the winding.

The radial dimension of the shoes also plays a role in cogging torque production. If the radial shoe dimension, *e.g.*, d_{sh} in Fig. 4-29, becomes too small, the ferromagnetic shoe tips become saturated by the magnet flux, thereby adding another varying component to the reluctance that produces cogging torque. In general, the radial shoe dimension is determined by manufacturing considerations. When stator laminations are cut or stamped it is difficult to form areas having width less than one to two times the lamination thickness. Therefore, the lamination thickness places a lower limit on the shoe radial dimension.

Since reluctance is given by $R=l/\mu A$, reluctance is directly proportional to length l . This fact leads to a more subtle way to minimize cogging torque. Over the stator teeth the flux path length is equal to the air gap length g , whereas over the stator slots the length is $g+l_s$ where l_s is the additional flux path length experienced by flux entering the slot opening. Using the circular arc, straight line flux path approximation developed in Chapter 2, l_s has a maximum of approximately $l_{s-max}=\pi w_{so}/4$, where w_{so} is the slot opening. Using this approximation, the flux path length from the magnet to the stator varies from g to $g+\pi w_{so}/4$. Therefore, minimizing this variation minimizes the reluctance variation and the resulting cogging torque. Minimizing the slot opening w_{so} minimizes this flux path length variation and minimizes cogging torque. In addition, increasing the air gap length g relative to the slot opening decreases the flux path variation and decreases cogging torque. If the air gap length is increased, the magnet length must be increased by the same percentage to keep the permeance coefficient P_c and resulting air gap flux constant. If the air gap flux decreases, the cogging torque will decrease due to the drop in flux as well, but the desired mutual torque will diminish as well, leading to lower motor performance.

Since each magnet produces cogging torque as it passes by stator slots, the relationship between the number of magnet poles and the number of stator slots influences cogging torque. In integral slot motors such as the four pole, twelve slot motor shown in Fig. 4-39, each magnet appears in the same position relative to the stator slots. As a result, the cogging torque created by all magnets are in phase with each other, and the net cogging torque is equal to the product of the number of magnet poles and the cogging torque created by one magnet. That is, the cogging torque from each magnet simply adds to create the net result.

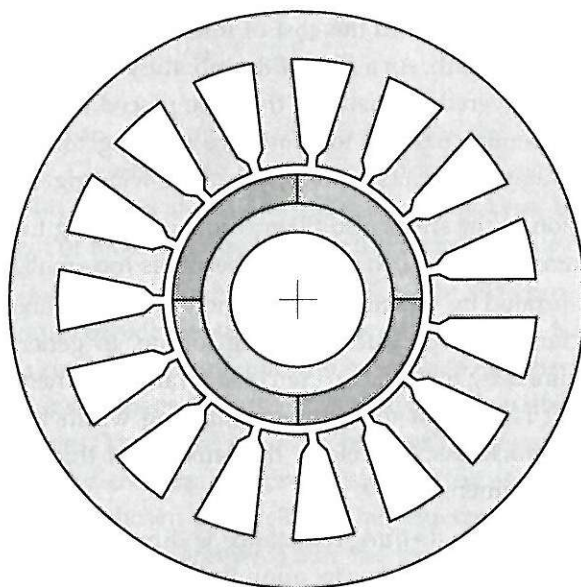


Figure 4-40. Cross section of a four pole, fifteen slot motor.

On the other hand, in fractional slot motors such as the four pole, fifteen slot motor shown in Fig. 4-40, each magnet appears in a different position relative to the stator slots. As a result, the cogging torques created by all magnets are out of phase with each other, and the net cogging torque is reduced since the cogging torque from each magnet adds together and at least partially cancels the cogging torque from other magnets. This fact is one of the primary reasons for choosing a fractional slot motor.

The final fundamental way to decrease cogging torque is also based on the $dR/d\theta$ term in (4.44). The reluctance R describes reluctance that extends along the axial direction of the motor as well. From this point of view, the net change in reluctance can be minimized, despite the slot openings, if the slot openings are spread out over the surface area of the magnet as depicted in Fig. 4-41. Here, the slots are skewed so that each magnet sees a net reluctance that stays the same or nearly the same as slots pass by. In this way, changes along the axial dimension are used to diminish the effect of changes along the circumferential dimension. As a result, the $dR/d\theta$ experienced by the entire magnet decreases and the cogging torque decreases.

While the slots are shown skewed in Fig. 4-41, the phenomenon does not change if the magnets are skewed instead. In this case each magnet still sees a net reluctance that stays the same or nearly the same as a function of position.

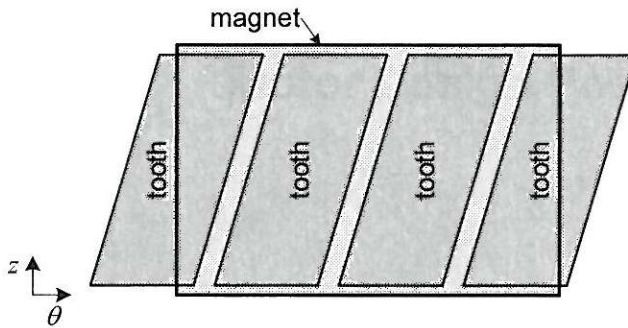


Figure 4-41. Skewed stator slots.

4.13 Summary

The fundamentals of brushless permanent magnet motors were discussed in this chapter. A magnetic circuit model for a motor was developed using simplifying assumptions that led to an expression for the air gap flux density. Based on an ideal flux density distribution, the flux linkage, back EMF, and torque produced by a coil on the stator were found. The influence of design variations evolved from this basic analysis. Since the stator windings place such a crucial role, coil properties of resistance and inductance were computed. In addition, discussion included the influence of coil connections and slot constraints. The key parameters of torque constant, back EMF constant, motor constant and torque per unit rotor volume were introduced. Finally, the chapter concluded with a discussion of cogging torque and design guidelines for its minimization.